Domestic vs. Foreign Law: Portfolio Dynamics of Sovereign Debt*

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Abstract

Emerging-market governments borrow both under their own law and under foreign law, usually that of New York or England. If the government defaults, the governing law determines whether domestic or foreign courts settle disputes, and so how much investors recover. Using daily prices of the same government's domestic-law and foreign-law bonds, we find that investors prefer foreign-law debt when default becomes more likely. Although foreign-law debt protects investors in default, all the governments we study also issue domestic-law debt, at shorter durations. We explain these duration differences with the fee on foreign-law bonds, which is proportional to face value and does not depend on maturity. We build a quantitative model of sovereign default in which the government can commit neither to repay nor to limit its future borrowing. Calibrated to Colombia, the model implies that investors expect to recover 47 percent of what they are owed under foreign law and 40 percent under domestic law. Removing the fee, which lets the government issue short foreign-law debt, raises welfare by 12 percent of the gain from eliminating aggregate risk and default costs.

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1 Introduction

When a government defaults, its creditors cannot seize assets within its borders, and what they can seize abroad generally falls short of what they are owed. What they recover is instead negotiated with the government, which remains excluded from financial markets until the two sides settle (Eaton and Gersovitz, 1981; Aguiar and Gopinath, 2006; Arellano, 2008; Yue, 2010). The bonds' terms set the rules of that negotiation; for example, they may set the majority of bondholders that must accept a deal for it to bind all of them. Whether the government can change those terms depends on the bond's governing law, the law under which they are written and enforced. Under domestic law, the government can change them as it would any other law, even after the bonds have been sold. Under foreign law, a court outside the government's reach enforces them as written at issuance. In its 2012 restructuring, Greece used this difference to gain an advantage in the negotiation: it changed the terms of its Greek-law bonds but could not do so for its foreign-law bonds, and foreign-law bondholders recovered more on average (Zettelmeyer et al., 2013).

This paper studies how governments borrow under domestic and foreign law, empirically and quantitatively. The central object is how much investors expect to recover under each law. Estimates from past restructurings vary widely across countries and episodes (Cruces and Trebesch, 2013), and say little about what investors expect from a government that has not defaulted recently. We therefore develop a methodology that infers the expected recovery under each law from a country's own bond prices and the amount of debt it issues under each law, before any default occurs. We apply it to Colombia and find that investors expect to recover 40 percent of their claims under domestic law and 47 percent under foreign law. We then explain why Colombia issues under both laws, and measure what it would gain if its domestic-law debt offered investors the same protection in default as its foreign-law debt.

We begin by documenting how emerging-market governments use the two laws. We combine debt-office records with bond-level data from Dealogic on every bond twelve governments issued between 2006 and 2024, and with daily bond prices. Three facts emerge. First, all twelve governments issue under both laws, with foreign-law shares of marketable debt between 5 percent in Brazil and 63 percent in Uruguay. Second, foreign-law debt is issued at longer duration in every country. Third, the yield gap between the two laws widens with default risk. To measure it, we price each foreign-law bond's payments on the domestic-law yield curve, following Chamon et al. (2018), and use bonds that Brazil, Colombia, Paraguay and Russia issue in their own currency under foreign law,

so that the two bonds differ only in their governing law. In Colombia, the gap rises by 23 basis points for every 100 basis points of CDS.

We then develop a quantitative model of sovereign default in which a government that lacks commitment chooses how much to borrow under each law (Arellano, 2008; Chatterjee and Eyigungor, 2012). Debt under each law has its own recovery and its own duration. Calibrated to Colombia, with durations of 3.75 years under domestic law and 8.71 years under foreign law, the model needs two moments to pin down the two recoveries: the slope of the yield gap and the share of debt under foreign law.

The model explains why a government that lacks commitment issues under both laws. The government can commit neither to repay nor to a path of future borrowing. On the one hand, foreign-law debt recovers more, which protects investors against default. On the other hand, domestic-law debt is shorter, which protects them against dilution: new borrowing lowers the value of debt that is still outstanding, and long debt stays exposed for longer (Hatchondo et al., 2016; Aguiar et al., 2019). Each law addresses one of the two commitment problems, so the government uses both. With equal durations, it issues only the high-recovery bond; with equal recoveries, only the shorter bond.

We then ask why governments do not issue short foreign-law bonds, which would combine the higher recovery of foreign law with protection against dilution. Foreign-law bonds are placed through bank syndicates for a fee that is a fixed share of face value, independent of the bond's maturity and of the amount issued (Section 2.5), while domestic-law bonds are auctioned at almost no cost. The fee is a tax on rolling over debt, so it weighs most on short bonds. With only one-period bonds, a fee the size of the one we measure barely changes the government's choice. With a long foreign-law bond available, the same fee reduces short foreign-law debt from 6.7 to 0.5 percent of GDP, because the government can offer investors the recovery of foreign law through the long bond, on which it pays the fee less often.

Last, we use the model to measure what the fee costs the government. Without the fee, the government would replace its short domestic-law bond with a short foreign-law bond, which has the same duration and a higher recovery. In the model, this is achieved by raising the recovery of the short bond from 40 to 47 percent. Welfare rises by 8.2 basis points of permanent consumption, about 12 percent of the gain the baseline model generates when aggregate risk and default costs are removed. The government borrows more, by 0.4 percentage points of GDP, defaults less often, by 0.3 percentage points a year, and spreads fall by 46 basis points on short debt and 22 basis points on long debt. Decompos-

ing the gain, we find that most of it comes from changes in the government's borrowing and default policies, and a smaller part from lower borrowing costs, the channel the literature on governing law emphasizes (Chamon et al., 2018). The result is robust to positive initial levels of debt. As is standard in quantitative models of sovereign default, the government is more impatient than investors. Its impatience may reflect political frictions more than the preferences of households (Aguiar et al., 2020). We therefore also evaluate the gain with a higher household discount factor, and households more patient than the government also gain.

Related literature. Chamon et al. (2018) show that in the euro area the yield gap between foreign-law and domestic-law bonds rises with default risk. Most foreign-law bonds are issued in a different currency from domestic-law bonds, so their main estimates convert currencies. In emerging markets, defaults come with large depreciations, which prevents us from following this approach. Saiegh and Biglaiser (2024) use identical dollar bonds that Argentina issued under both laws in 2020, and find that the foreign-law bonds trade at higher prices. Such twins are rare. We compare local-currency bonds issued under both laws, which several emerging markets have. We study four of them and give the first estimates for Brazil, Colombia and Paraguay. In all four, the gap rises with default risk.

Estimates of how much creditors recover in default come from past restructurings, mostly of external debt (Cruces and Trebesch, 2013). These episodes are rare for any one government, and their recoveries vary widely. Domestic law governs between 65 and 95 percent of the marketable debt in ten of the twelve governments we study, and evidence on defaults on this debt is recent (Erce et al., 2022). Arellano et al. (2023) and Arellano and Barreto (2025) estimate recoveries by matching a quantitative model of default to the data. They treat the recovery on each type of debt as a fixed parameter, which summarizes the outcome of the negotiation between the government and its creditors. We follow this approach for the two laws, taking the durations from the data and matching the share of debt under each law and how the yield gap moves with default risk. (See also Sturzenegger and Zettelmeyer 2008 and Graf von Luckner et al. 2025 on haircuts; Yue 2010, Benjamin and Wright 2009 and Asonuma and Trebesch 2016 on negotiation; and Schumacher et al. 2018 and Fang et al. 2021 on litigation and holdouts under foreign law.)

We extend the quantitative models of sovereign default of Aguiar and Gopinath (2006) and Arellano (2008) by letting the government choose the law under which it borrows. As

in Chatterjee and Eyigungor (2012) and Hatchondo et al. (2016), the government cannot commit to its future borrowing, so new debt dilutes the value of long-term debt. Several papers let the government choose among types of debt that serve different purposes. In Arellano and Ramanarayanan (2012), short-term debt gives incentives to repay, and long-term debt hedges against changes in spreads. In Ottonello and Perez (2019), foreign-currency debt avoids the costs of inflation, and local-currency debt provides a hedge. Choi et al. (2026) let the government choose currency and maturity together, and Arellano and Barreto (2025) let it default selectively on official and private lenders. In our model, the same investors hold both bonds, and the government defaults on both at once. The two bonds differ in the law that governs them, which sets how much investors recover, and in their duration, which we take from the data. (See also Eaton and Gersovitz 1981, Hatchondo and Martinez 2009, Du et al. 2020 and Passadore et al. 2026. As in Passadore and Xu 2022, a government that returns to the market still carries part of its defaulted debt; Yue 2010 and Mihalache 2020 model how that debt is renegotiated.)

We also contribute to work on the maturity of sovereign debt. A government that cannot commit to its future borrowing has a reason to borrow short, because short-term debt is less exposed to dilution (Hatchondo et al., 2016). A cost of issuing works the other way: long debt is rolled over less often, so it pays the cost less often (Jungherr et al., 2024). In Bigio et al. (2022), the cost comes from auctions in which larger issues fetch lower prices, and the government spreads its issuance across maturities. The cost we study is the fee that banks charge to place foreign-law bonds, which keeps the government from issuing short foreign-law bonds. In a simple model without default, we show that with a fee proportional to the amount issued the government issues only the longest bond (Appendix A.20). We measure what the fee costs the government using the welfare decomposition of Aguiar et al. (2020). (See also Missale and Blanchard 1994, Jeanne 2009, Broner et al. 2013 and Aguiar et al. 2019 on maturity; and Lee et al. 1996, Altınkılıç and Hansen 2000 and Melnik and Nissim 2003 on underwriting fees.)

2 Empirical Evidence

2.1 Data

Our sample covers the central governments of twelve emerging markets from 2006 to 2024: Brazil, Chile, Colombia, Hungary, Mexico, Paraguay, Peru, Poland, Russia, South Africa, Turkey and Uruguay. We study their marketable debt: central-government bonds that any private investor can buy and later resell, net of the central bank's holdings. This

definition excludes four kinds of public debt. The first is loans from multilateral and bilateral lenders. The second is bills issued by the central bank and the debt of state companies. The third is retail savings bonds. The fourth is bonds sold directly to public institutions, such as state pension funds or development banks, without an auction. These institutions are often required by law to hold government debt, a form of financial repression (Reinhart and Sbrancia, 2015; Chari et al., 2020), so their purchases do not reflect the price at which private investors would lend. Bonds that public institutions buy at the government's regular auctions remain in the sample.

Domestic-law bonds come from each debt office's records, which report for every placement its date, amount, coupon, maturity and yield at issuance. Foreign-law bonds come from Dealogic's bond-level issuance data, which report the same variables together with the governing law. On both sides we keep only bonds the central government itself issues, and we treat each re-tap of an outstanding bond as a separate placement. Daily bond prices come from Bloomberg, official sources and Capital IQ Pro. When more than one source covers a bond, we use the source with the widest coverage and check the results against the others. Daily five-year CDS spreads come from Markit, through Capital IQ Pro. Underwriting fees come from Dealogic, and the issuer's own expenses from the final prospectuses that governments file with the SEC. GDP in current US dollars comes from the World Bank's World Development Indicators.

2.2 Every government borrows under both laws

Every government in our sample borrows under both laws. Table 1 reports each government's marketable debt under each law, as a percentage of GDP and as a share of its total marketable debt, averaged over 2006–2024. Foreign-law bonds make up between 5 percent of marketable debt in Brazil and 63 percent in Uruguay, and between 20 and 35 percent in seven of the twelve countries. As a percentage of GDP, foreign-law debt ranges from 1.5 percent in Russia to 20 percent in Uruguay, and total marketable debt from 7 percent in Paraguay to 48 percent in Hungary.

We measure the stock of marketable debt under each law at the end of each year from 2006 to 2024. A bond counts as outstanding from its settlement date until it matures. The domestic-law stock comes from the debt offices' records, with inflation-indexed bonds valued at what the government owes on them, and we convert it to dollars at the year's average exchange rate. The foreign-law stock is the dollar value of each bond at issuance, from Dealogic. Appendix A.1 reports the shares under other exchange-rate conventions, with foreign-law bonds revalued each year, and without the holdings of all public insti-

Table 1. Marketable Debt by Governing Law

	Domestic law		Foreign law		Total
	% GDP	% debt	% GDP	% debt	% GDP
	(1)	(2)	(3)	(4)	(5)
Brazil	41.9	95.0	2.2	5.0	44.1
Chile	14.5	76.2	4.7	23.8	19.2
Colombia	14.3	70.3	6.1	29.7	20.4
Hungary	33.6	69.8	14.7	30.2	48.3
Mexico	23.5	84.0	4.5	16.0	28.0
Paraguay	1.2	48.1	5.6	51.9	6.8
Peru	10.2	65.3	5.2	34.7	15.4
Poland	29.6	79.2	7.8	20.8	37.4
Russia	6.5	79.1	1.5	20.9	8.0
South Africa	41.4	91.5	3.9	8.5	45.3
Turkey	19.2	75.5	5.5	24.5	24.7
Uruguay	11.8	37.5	20.1	62.5	31.9

Notes: Marketable central-government debt outstanding at each 31 December, averaged over 2006–2024. Marketable debt consists of the central-government bonds that any private investor can buy on the same terms and later resell, net of the central bank's holdings. Columns (1) and (3) report the domestic-law and foreign-law stocks as a percentage of GDP, columns (2) and (4) as a percentage of total marketable debt, and column (5) the total as a percentage of GDP. Debt and GDP are in current US dollars, GDP from the World Bank (NY.GDP.MKTP.CD). Domestic-law stocks come from each debt office's records and foreign-law stocks from Dealogic. Robustness checks on column (4) are in Table 10.

tutions.

2.3 Foreign-law debt is issued at longer duration

Table 2. Duration at Issuance by Governing Law

	Domestic law				Foreign law				Gap	
	Median (1)	Mean (2)	Face (3)	GDP (4)	Median (5)	Mean (6)	Face (7)	GDP (8)	Median (9)	Face (10)
Brazil	3.47	4.35	2.34	2.28	8.40	9.65	9.87	9.82	4.93	7.53
Chile	8.85	10.31	8.36	8.57	9.90	12.13	12.35	12.26	1.05	3.99
Colombia	3.75	4.74	5.79	5.52	8.71	10.56	10.58	10.62	4.96	4.79
Hungary	2.84	3.74	2.46	2.35	7.54	7.60	8.45	8.39	4.70	5.99
Mexico	0.48	2.67	2.00	1.96	8.87	10.14	11.12	11.13	8.39	9.12
Paraguay	3.46	3.88	4.08	3.87	8.74	10.54	10.51	10.54	5.28	6.43
Peru	8.64	9.28	9.10	9.30	11.77	13.46	13.26	12.97	3.13	4.16
Poland	2.56	3.97	3.72	3.63	8.19	8.75	8.80	8.80	5.63	5.08
Russia	5.80	5.49	4.54	4.64	8.00	8.98	9.51	9.55	2.20	4.97
South Africa	6.88	6.89	3.57	3.59	8.79	10.12	10.23	10.16	1.91	6.66
Turkey	1.43	2.25	2.13	1.86	7.64	7.83	7.43	7.57	6.21	5.30
Uruguay	8.19	8.55	8.91	8.78	9.45	10.55	11.51	11.47	1.26	2.60

Notes: Macaulay duration of each placement at the moment of sale, in years. Median and Mean count every placement once; Face weights each placement by the dollars it raised; GDP weights it by those dollars over its own year's GDP. Gap is foreign law minus domestic law. Samples start in 2004 or 2006 and end in 2024. Maturity at issuance, a common-window check and the governing law of foreign-law placements are in Tables 11, 12 and 13.

Foreign-law debt is issued at longer duration than domestic-law debt in every country in our sample. Table 2 reports the duration at issuance of each government's placements under each law. The median foreign-law placement has a duration between 7.5 years in Hungary and 11.8 years in Peru, and the median domestic-law placement between 0.5 years in Mexico and 8.9 years in Chile. The gap between the two medians ranges from 1.1 years in Chile to 8.4 years in Mexico. Figure 1 plots the distribution of duration at issuance under each law. In most countries, domestic-law placements concentrate at short durations, where foreign-law placements are rare.

We measure the duration of every placement of marketable debt at the time of sale, following Macaulay (1938): the average of the years until each of the bond's payments, each weighted by its discounted value relative to the bond's price. For a bond with face value of 1 that pays coupon c at $n_1, \ldots, n_J$ years after settlement and is sold at yield y,

$$d = \frac{1}{p}\left[\sum_{j=1}^{J} \frac{n_j\, c}{(1+y)^{n_j}} + \frac{n_J}{(1+y)^{n_J}}\right], \qquad p = \sum_{j=1}^{J} \frac{c}{(1+y)^{n_j}} + \frac{1}{(1+y)^{n_J}}.$$

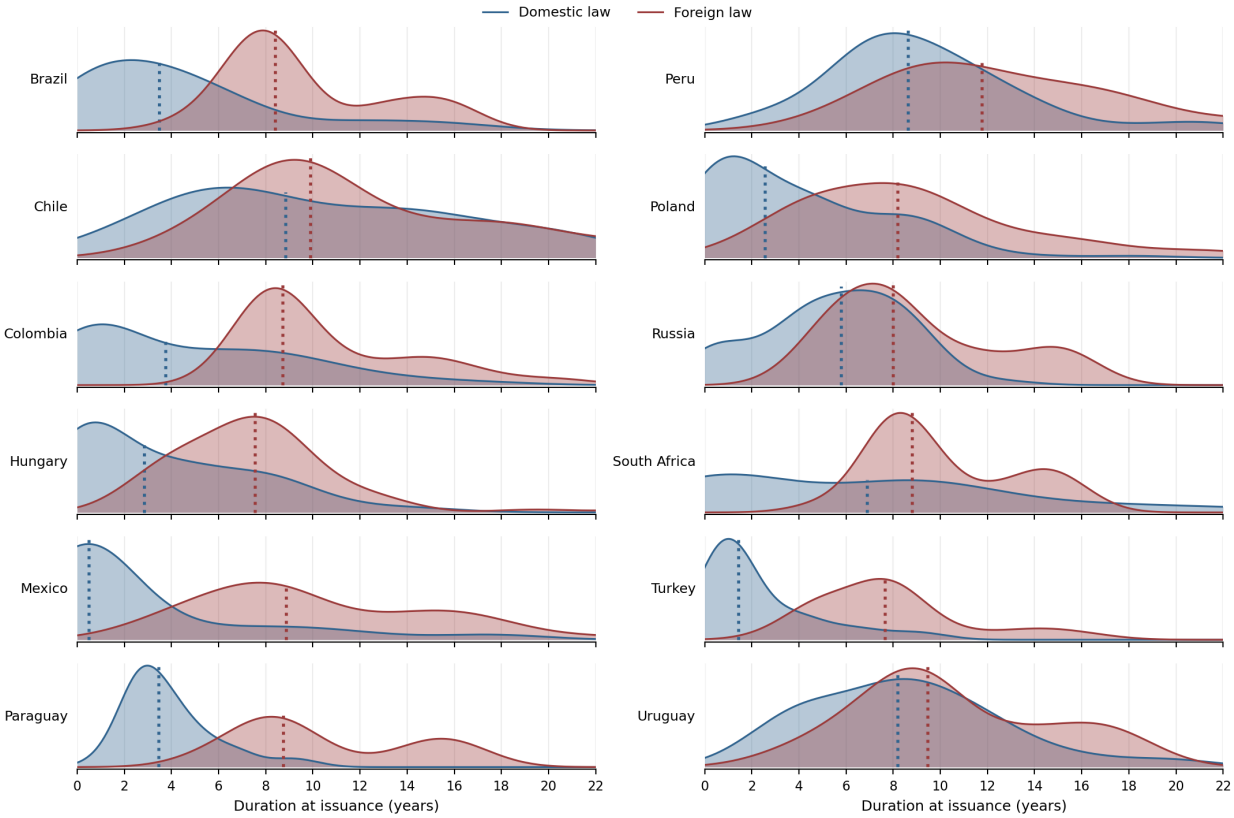


Figure 1. Distribution of Duration at Issuance by Governing Law

Notes: Kernel densities of duration at issuance, in years, over each government's placements under each law, 2006–2024 (2004–2024 for Hungary, Poland, Russia, South Africa and Turkey). Each placement, including each re-opening, counts once. Dotted lines are the medians of Table 2. The horizontal axis stops at 22 years; 0.8 percent of placements lie beyond it and enter the estimates.

As in Arellano and Ramanarayanan (2012), we use duration because it is comparable across bonds with different coupons. A re-tap of an outstanding bond counts as a separate placement, measured from its own settlement date. For each law we report the median and mean across placements, and two weighted means: one weights each placement by the dollars it raised, and the other by those dollars over its year's GDP. Appendix A.2 describes how we treat each type of bond, and reports maturity at issuance, the medians over a common 2006–2024 window, and the laws under which the foreign-law placements are written.

2.4 The yield gap rises with the probability of default

The yield gap between domestic-law and foreign-law bonds of the same government rises with the probability of default. We define the gap as the domestic-law yield minus the foreign-law yield. If investors expect to recover more under foreign law, that extra recovery is worth more when default is more likely, so the gap widens as default becomes more likely. If investors do not care which court enforces the bond, the gap reflects only the other characteristics of the two bonds and does not move with default risk (Appendix A.3 derives this relation in a simple pricing model). The yields of a government's domestic-law and foreign-law bonds cannot be compared directly, because the bonds differ in duration (Section 2.3), coupon and, in most cases, currency. We therefore use the foreign-law bonds that Brazil, Colombia, Paraguay and Russia issue in their own currency (Table 3). As in Chamon et al. (2018), we construct for each of them a domestic-law bond with the same payments. Because both bonds pay in the same currency, we need no currency conversion, which in emerging markets would mix the fall of the currency in default with the value of the law (Appendix A.3).

Table 3. Local-Currency Foreign-Law Bonds

Country	Law	Currency	Bonds	Instruments	Window
Brazil	New York	BRL	3	Global BRL 12.5% '22, 10.25% '28, 8.5% '24	2014–2020
Colombia	New York	COP	3	Global TES 9.85% '27, 7.75% '21, 4.375% '23	2017–2020
Paraguay	New York	PYG	3	Global 7.9% '31, 8.5% '35, 8.5% '38	2025–2026
Russia	English	RUB	1	7.85% '18 eurobond	2014–2017

Notes: All bonds are nominal, fixed-coupon and non-callable. Brazil's and Colombia's windows end at the COVID break, Russia's when its bond falls below one year to maturity, and Paraguay's at the end of available data.

The construction proceeds in four steps. A foreign-law bond has a daily price $q_{F,t}^{LC}$ and payments $c_1, \ldots, c_n$ per unit of face on dates $s_1 < \cdots < s_n$, the last one including princi-

pal. First, we compute its yield to maturity $y_{F,t}$ from its price,

$$q^{LC}_{F,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + y_{F,t}\right)^{s_k - t}}. \tag{1}$$

Second, we fit the domestic-law zero-coupon yield curve $z_t(\cdot)$ on day t, and discount the same payments on it,

$$\widetilde{q}^{LC}_{D,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + z_t(s_k - t)\right)^{s_k - t}}. \tag{2}$$

Third, we compute the yield $\widetilde{y}_{D,t}$ that this price implies,

$$\widetilde{q}^{LC}_{D,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + \widetilde{y}_{D,t}\right)^{s_k - t}}. \tag{3}$$

Fourth, the yield gap is $\widetilde{y}_{D,t} - y_{F,t}$. The two bonds then share issuer, currency and payments, and differ only in their governing law. The domestic-law curve has to contain only domestic-law bonds. We use published curves where they do: Bloomberg's for Brazil and Colombia and the central bank's for Russia. For Paraguay, where no such curve exists, we fit a Nelson–Siegel curve each day to the domestic-law bonds quoted that day (Appendix A.7).

We measure the probability of default with the government's five-year CDS spread, and estimate how the gap moves with it, country by country, in first differences:

$$\Delta\left(\widetilde{y}_{D,i,t} - y_{F,i,t}\right) = \gamma\, \Delta \mathrm{CDS}_t + \Gamma' X_{i,t} + \mu_i + u_{i,t}, \tag{4}$$

where i indexes bonds and t trading days. The controls $X_{i,t}$ are the change in the bond's bid-ask spread relative to the domestic benchmark, which accounts for differences in trading costs; its time to maturity; and the change in the VIX, which separates movements in the CDS spread that reflect global risk aversion. The μ_i are bond fixed effects. The coefficient γ is the number of basis points the gap moves per basis point of CDS. We use first differences because we cannot reject a unit root in the gap or the CDS spread (Appendix A.8), which also removes any constant wedge between the two bonds, such as an error in fitting the curve. We use Driscoll–Kraay standard errors because the CDS spread is common to all of a country's bonds.

The gap rises with default risk in every country (Table 4). For every 100 basis point rise in the CDS spread, it rises by 69 basis points in Brazil, 23 in Colombia, 17 in Paraguay and 7 in Russia, and each estimate is significant at the 5 percent level or better. Because the

Table 4. The Yield Gap and the CDS Spread

	Brazil	Colombia	Paraguay	Russia
Δ CDS spread	0.694***	0.225***	0.169**	0.073***
	(0.102)	(0.059)	(0.078)	(0.026)
	[0.071]	[0.047]	[0.080]	[0.032]
Liquidity, maturity, VIX controls	Yes	Yes	Yes	Yes
Bond fixed effects	Yes	Yes	Yes	—
Bonds	3	3	3	1
Observations	3,332	1,347	713	633
R^2	0.098	0.036	0.043	0.078
Window	2014–2020	2017–2020	2025–2026	2014–2017

Notes: Estimates of γ in equation (4), country by country. The dependent variable is the daily change in the yield gap, in basis points; ΔCDS is the daily change in the five-year CDS spread. Controls: the change in the bond's bid-ask spread relative to the domestic benchmark, time to maturity and the change in the VIX, with bond fixed effects. Driscoll–Kraay standard errors in parentheses (Newey–West for Russia, which has one bond), Huber–White in brackets. Stars follow Driscoll–Kraay: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

local currency also loses value when the government defaults, these slopes understate the value of foreign law to a dollar investor; Appendix A.4 measures this effect.

Reading the gap as the value of governing law requires that the same investors hold both bonds. Each country's sample therefore starts once nonresidents can buy, hold and sell the domestic-law bond without restrictions or punitive taxes, and over these windows nonresidents held on average 15 percent of the domestic-law debt in Brazil, 26 percent in Colombia and 23 percent in Russia (Appendix A.5). As a placebo, we apply the same construction to Peru's sol Global Depositary Notes, which are held offshore like the other bonds but wrap domestic-law bonds, so the governing law is the same on both sides; their gap does not move with Peru's CDS spread (Appendix A.11). Appendix A.9 reports estimates under alternative specifications, windows and samples of bonds.

2.5 Issuing under foreign law costs a fee proportional to face value

Issuing under foreign law costs the government a fee proportional to the face value of the bonds it sells, at a rate that does not depend on the maturity of the bonds or on the amount issued. A government that sells a foreign-law bond hires a syndicate of banks, which buy the bonds and place them with investors, and the banks keep a share of face value, the underwriting fee. Domestic-law bonds are sold at auctions run by the central

bank or the debt office, which charge no comparable fee. Because the fee is paid each time a bond is issued, a government that borrows short under foreign law pays it more often for the same debt. Section 5.1 uses this fact to explain why governments issue foreign-law bonds at longer duration, as we document in Section 2.3.

We measure the fee with Dealogic's gross spread, the underwriting fee of each bond in basis points of face value. We restrict the sample to bonds issued under New York law, for which Dealogic records a fee for 92 percent of the bonds, against 9 percent of English-law bonds and 43 percent of Japanese-law bonds. The sample has 368 bonds issued by ten governments between 2006 and 2024; Dealogic records no fee for Paraguay and Russia. Appendix A.12 repeats the analysis for the bonds issued under other foreign laws: bonds issued under English law in euros pay about the same fee as New York-law bonds, 14 against 15 basis points, while bonds issued in yen and Swiss francs pay more.

Table 5. Underwriting Fee on Foreign-Law Bonds

	Bonds	Median	Mean	Mean, face-weighted
	(1)	(2)	(3)	(4)
Brazil	50	25.0	24.4	22.9
Chile	47	4.0	5.6	5.3
Colombia	49	25.0	27.7	25.7
Hungary	10	18.0	19.1	18.2
Mexico	72	19.0	22.2	19.3
Peru	23	4.0	8.1	7.1
Poland	15	14.0	14.8	15.2
South Africa	19	13.0	13.8	13.7
Turkey	56	8.0	8.2	7.9
Uruguay	27	10.0	11.6	10.7

Notes: Underwriting fee (gross spread) in basis points of face value, from Dealogic, New York-law bonds, 2006–2024, one observation per bond, re-openings included. Columns (2) and (3) count every bond once; column (4) weights each bond by its face value in US dollars. Dealogic records no fee for Paraguay and Russia.

Each government pays its own rate (Table 5). The median fee ranges from 4 basis points in Chile and Peru to 25 in Brazil and Colombia. Figure 2 plots the fee of each bond against its maturity and against its size, after removing country and year averages.

The rate does not depend on the maturity of the bond or on the amount issued. We regress the fee on maturity and size, comparing bonds of the same country and year, and find no relationship with either. Neither coefficient is significant, and we cannot reject that both are zero ($p = 0.82$; Table 6, Figure 2, Appendix A.12). Each government

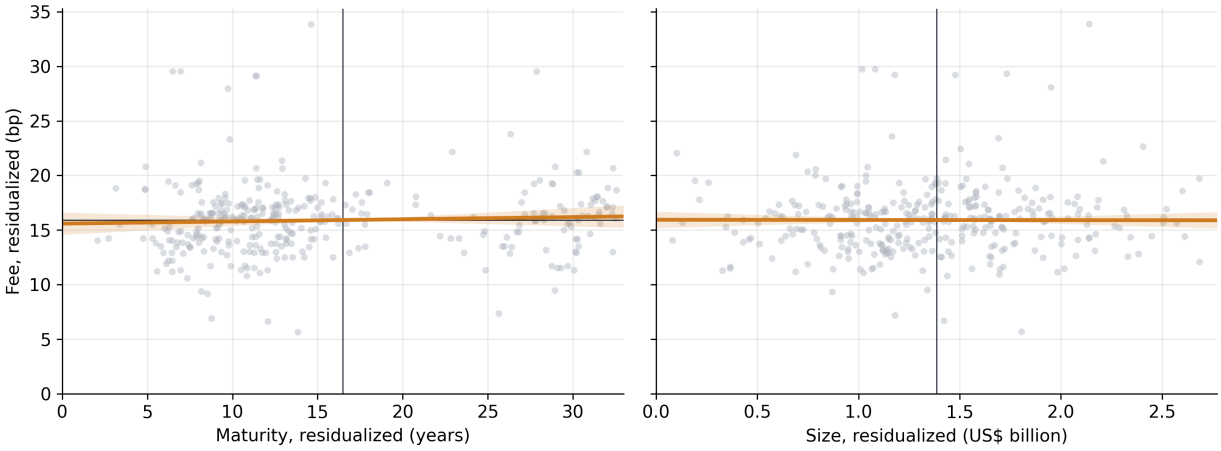


Figure 2. Underwriting Fee, Maturity and Size

Notes: Underwriting fee of each New York-law bond, in basis points of face value, against its maturity at issuance (left) and its face value in US$ billion (right), 2006–2024, 360 bonds. Each axis is the residual from a regression of that variable on country and year effects and on the other regressor, with the sample average added back (fee 15.9 basis points, maturity 16.5 years, size US$1.4 billion). The line is the least-squares fit, and the shaded band is its 95 percent interval, with errors clustered by country. Bonds beyond twice the average on either axis are in the regression but not drawn.

Table 6. Regression of the Underwriting Fee on Maturity and Size

	Fee (bp of face)
Maturity (years)	0.020
	(0.032)
	[0.61]
Size (US$ billion)	−0.016
	(0.276)
	[0.95]
Country effects	Yes
Year effects	Yes
Countries (clusters)	10
Bonds	360
R^2	0.871

Notes: OLS regression of the underwriting fee of each New York-law bond, in basis points of face value, on its maturity at issuance in years and its face value in US$ billion, with country and year effects, 2006–2024. The sample excludes two retail notes that Mexico sold in 2006 and the bonds below the 1st and above the 99th percentile of maturity. Standard errors clustered by country in parentheses; wild-bootstrap p-values in brackets. The joint test that both coefficients are zero gives $p = 0.82$.

therefore faces a single rate per unit of face value, whatever the size or maturity of the bonds it issues. Earlier evidence on government bonds points the same way. Among the 37 government and agency bonds placed in 1996–97 that Melnik and Nissim (2003) study, the fee is unrelated to maturity and to size. Among the emerging-market government bonds placed between 1993 and 2007, the fee does not rise with the spread at which the bond is sold (Flandreau et al., 2010), so it does not price the risk of each bond either.

The government also pays a flat amount for each deal, which is similar across countries and deals. These are its own expenses: lawyers, listing and paying agents, which the prospectuses filed with the SEC state as a single estimate. The median ranges from US$120,000 in Uruguay to US$305,000 in South Africa, a deal twice as large costs only 14 percent more in expenses, and expenses account for about 8 percent of the cost of a deal (Table 21).

Domestic auctions carry no comparable charge. In Brazil and Russia, the law states that the central bank runs them free of charge. Chile, Colombia, Turkey and Paraguay pay their central banks small flat amounts or reimburse their costs; in Chile, the fee for an issuance programme, relative to the size of the programme, is less than a thirtieth of the fee Chile pays on its foreign-law bonds. Uruguay is the exception: its central bank charges 1 basis point a year on the stock of domestic-law bonds. For the other countries we find no charge in the documents we reviewed.

3 Model

In this section, we present a long-term debt model of sovereign default with two assets and heterogeneous recovery rates after default. Section 3.1 describes the macroeconomic environment, section 3.2 describes the timing, 3.3 characterizes the decisions of the government given prices, section 3.4 defines bond prices and section 3.5 defines the equilibrium.

3.1 Small open Economy

Time is discrete and denoted by $t \in \{0, 1, 2, \ldots\}$. The small open economy receives a stochastic stream of income denoted by y_t. Income follows a first-order Markov process $\mathbb{P}(y_{t+1} = y' \mid y_t = y)$. The government is benevolent and trades foreign-law and domestic-law bonds to smooth the household's consumption. The household evaluates

consumption streams, c_t, according to:

$$\mathbb{E}\left[\sum_{t=0}^{\infty}\beta^t u\left(c_t\right)\right],$$

with time-preference $\beta \in (0,1)$ and utility function $u(\cdot)$, with $u'(\cdot) > 0$ and $u''(\cdot) < 0$.

The government issues foreign-law and domestic-law debt when it is not under default. As in Chatterjee and Eyigungor (2012) and Hatchondo and Martinez (2009) each unit of domestic (foreign) law debt matures with probability λ_D (λ_F). Bonds are issued at prices $q^{ND}_{D,t}$ and $q^{ND}_{F,t}$ at period t, and in equilibrium, those prices will depend on the current income and the vector of the next period's bond position $(y, b_{D,t+1}, b_{F,t+1})$. The government's budget constraint is:

$$c_t + \lambda_D b_{D,t} + \lambda_F b_{F,t} = y_t + q_{F,t}[b_{F,t+1} - (1-\lambda_F)b_{F,t}] + q_{D,t}[b_{D,t+1} - (1-\lambda_D)b_{D,t}],$$

where $\lambda_D b_{D,t}$ ($\lambda_F b_{F,t}$) are the total principal payments due corresponding to domestic-law (foreign-law) debt and $q_{j,t}[b_{j,t+1} - (1-\lambda_j)b_{j,t}]$ are the corresponding new debt issuances for debt under jurisdiction $j = D, F$.

3.2 Timing

The timing for the government is as follows and is summarized in Figure (3). There is limited enforcement of debt; thus, the sovereign defaults if it is optimal. When in default, the government loses access to the international lending market and incurs an output cost of $\phi(y)$ for each period it remains excluded. Consumption under exclusion c^D is exogenously determined by the budget constraint. During each period under exclusion, the government might regain access to credit markets with probability θ.

In our model, debt does not erase after default. This means that we have to keep track of debt prices during the default episode since defaulted debt has a positive recovery value. In a given period, before θ has been realized, domestic-law defaulted bonds are valued according to q^D_D, while foreign-law bonds are valued according to q^D_F.

Let $b_{t,j}$ represent the total amount of debt defaulted in the past under legislation j. A fraction $(1-\nu_j)$ of defaulted debt under legislation j is written off when the government regains access to credit markets. Thus, after a favorable draw of θ the recovered outstanding debt for the next period under each legislation becomes $\nu_j b_{t,j}$. As in Passadore and Xu (2022), debt is not erased in default, and the government regains access with the

recovered debt outstanding. Appendix A.14 describes how we implement recoveries.

Bondholders of defaulted debt of jurisdiction j receive a fraction $\nu_j < 1$ identical replacement bonds for each unit they hold during exclusion. After being allowed back to financial markets, the newly issued domestic-law bonds are evaluated at $q_D^{ND}\left(y, \nu_D b_{t,D}, \nu_F b_{t,F}\right)$ while their foreign-law counterparts are evaluated at $q_F^{ND}\left(y, \nu_D b_{t,D}, \nu_F b_{t,F}\right)$.

The government either defaults on both domestic-law and foreign-law debt or repays both. In the database of Erce et al. (2024), restricted to countries where foreign investors could buy bonds under both laws, most default episodes hit both classes of bonds (Appendix A.6).

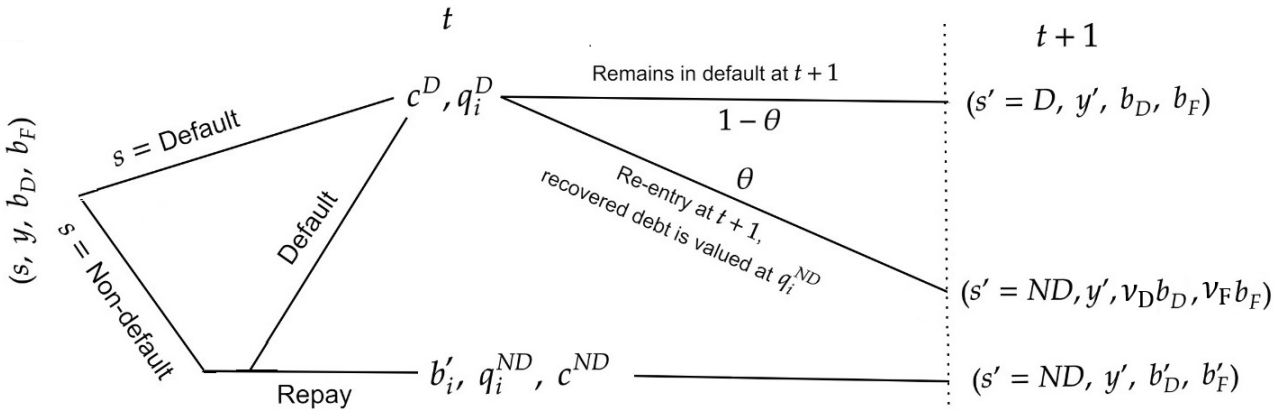


Figure 3. Timing

3.3 Recursive Formulations

Upon repayment, the government takes the price schedules as given and chooses a vector of bond holdings for the next period (b'_D, b'_F). We look for a Markov equilibrium in which strategies are conditioned on (y, b_D, b_F). The unconditional value of the government is given by:

$$V(y, b_D, b_F) = \max_{d \in \{0,1\}} dV^D(y, b_D, b_F) + (1-d)V^{ND}(y, b_D, b_F). \tag{5}$$

Where the d is the government's default policy:

$$d(y, b_D, b_F) = \mathbb{1}_{\left\{V^D(y, b_D, b_F) > V^{ND}(y, b_D, b_F)\right\}}. \tag{6}$$

In case of default, there is no choice in the government's value:

$$V^D(y, b_D, b_F) = u\left(y - \phi(y)\right) + \beta \mathbb{E}_{y'|y}\left[\theta V^{ND}\left(y', \nu_D b_D, \nu_F b_F\right) + (1-\theta)V^D\left(y', b_D, b_F\right)\right] \tag{7}$$

Under repayment, the government chooses the vector of new issuances b'_D, b'_F:

$$V^{ND}(y, b_D, b_F) = \max_{b'_D, b'_F} u(c) + \beta \mathbb{E}_{y'|y} \left[V^{ND}\left(y', b'_D, b'_F\right) \right] \tag{8}$$

Subject to the budget constraint and a bound in the default probability induced by the choice of debt:

$$\begin{aligned} c + \lambda_D b_D + \lambda_F b_F = y &+ q_F^{ND}\left(y, b'_D, b'_F\right)\left[b'_F - (1-\lambda_F) b_F\right] \\ &+ q_D^{ND}\left(y, b'_D, b'_F\right)\left[b'_D - (1-\lambda_D) b_D\right] \end{aligned} \tag{9}$$

$$\delta\left(y, b_D, b_F\right) \equiv \mathbb{E}_{y'|y}\left[d\left(y', b'_D, b'_F\right)\right] \leq \bar{\delta}. \tag{10}$$

As mentioned by Chatterjee and Eyigungor (2015) equation (10) is necessary for models with positive recovery rates to prevent the government from taking unrealistic amounts of debt right before defaulting. The solution to (8) subject to (9), and (10) generates the policy functions for foreign and domestic-law debt and an endogenous composition between the two.

3.4 Prices

Both bonds are priced by the same competitive, risk-neutral international investors, who discount payoffs at the risk-free rate r, as in Arellano (2008) and Chatterjee and Eyigungor (2012). Foreign investors hold a material share of the domestic-law debt of the governments we study (Appendix A.5). The value of one unit of debt when the government is not in default is given by:

$$\begin{aligned} q_D^{ND}\left(y, b'_D, b'_F\right) = \mathbb{E}_{y'|y} \Bigg\{ &\left(1 - d\left(y', b'_D, b'_F\right)\right) \frac{\lambda_D + (1-\lambda_D) q_D^{ND}\left(y', b''_D, b''_F\right)}{1+r} \\ &+ d\left(y', b'_D, b'_F\right) \frac{q_D^D\left(y', b'_D, b'_F\right)}{1+r} \Bigg\}, \end{aligned} \tag{11}$$

$$\begin{aligned} q_F^{ND}\left(y, b'_D, b'_F\right) = \mathbb{E}_{y'|y} \Bigg\{ &\left(1 - d\left(y', b'_D, b'_F\right)\right) \frac{\lambda_F + (1-\lambda_F) q_F^{ND}\left(y', b''_D, b''_F\right)}{1+r} \\ &+ d\left(y', b'_D, b'_F\right) \frac{q_F^D\left(y', b'_D, b'_F\right)}{1+r} \Bigg\}. \end{aligned} \tag{12}$$

When not in default, the prices reflect the expected payoff of the lender, discounted under the risk-free rate. In particular, note that since debt lasts more than one period, the price incorporates the optimal behavior of the government in the future, which we denoted as b_j''. Conditional on repayment tomorrow, the lender gets the principal for the matured fraction of each unit of debt, λ_j, plus tomorrow's price for the non-matured fraction, $1-\lambda_j$.

Note that in case of default, the price of one unit of debt is given by:

$$q_D^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\mathbb{E}_{y'|y}\left[q_D^D\left(y', b_D, b_F\right)\right] + \theta\nu_D q_D^{ND}\left(y, \nu_D b_D, \nu_F b_F\right), \tag{13}$$

$$q_F^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\mathbb{E}_{y'|y}\left[q_F^D\left(y', b_D, b_F\right)\right] + \theta\nu_F q_F^{ND}\left(y, \nu_D b_D, \nu_F b_F\right). \tag{14}$$

Consistent with Figure (3) defaulted debt prices reflect the probability of re-entering the market in the next period with probability θ. If the country receives a favorable draw, the recovered debt due is priced accordingly. Note that stated in this way, we allow for the possibility that the government defaults on the recovered debt, a feature not present in models where debt gets erased upon re-entry. On the other hand, if the country remains excluded from financial markets ($1-\theta$), it will face the same problem as today but possibly with a new income.

3.5 Equilibrium

Definition 1. *A Markov Equilibrium with state variables (y, b_D, b_F) is given by:*

(i) A set of policy functions for consumption $\hat{c}(y, b_D, b_F)$, default $\hat{d}(y, b_D, b_F)$, debt $\hat{b}'_D(y, b_D, b_F)$ and $\hat{b}'_F(y, b_D, b_F)$;

(ii) A set of value functions $\hat{V}(y, b_D, b_F)$, $\hat{V}^D(y, b_D, b_F)$, $\hat{V}^{ND}(y, b_D, b_F)$;

(iii) Price schedules $\hat{q}_D^D(y, b_D, b_F), \hat{q}_F^D(y, b_D, b_F), \hat{q}_D^{ND}(y, b'_D, b'_F), \hat{q}_F^{ND}(y, b'_D, b'_F)$

such that:

1. *Given prices, the value functions solve: (5), (7), and (8), with associated policy functions;*
2. *Bond prices satisfy (11), (12), (13), (14).*

4 Quantitative Analysis

4.1 Calibration

We calibrate the model to Colombia at an annual frequency. Colombia offers the most complete data for our exercise: its debt office records every domestic-law placement, and its foreign-law bonds are filed with the SEC, so the shares, durations, yield gap and fee we target all come from its own bonds. It is also a typical member of our sample. Its foreign-law share, 29 percent, lies within the range of seven of the twelve countries, and the slope of its yield gap lies between those of Russia and Brazil. Finally, Colombia has not defaulted since the 1930s, so the recoveries its investors expect cannot be read from its own restructurings and have to be inferred from prices, which is what our approach does. We solve a discretized version of the model with the algorithm of Gordon and Guerron-Quintana (2018), which extends Chatterjee and Eyigungor (2012) to models with several defaultable long-term bonds, building on the replication files of Deng et al. (2023) and Passadore and Xu (2022).

We set seven parameters outside the model. Household utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$, with relative risk aversion $\sigma = 2$, the standard value in the literature. Log output follows an AR(1),

$$\log y_t = \rho \log y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim N(0, \sigma_\varepsilon^2), \tag{15}$$

which we estimate on Colombia's annual real GDP from the World Development Indicators over 1976–2024, after removing a linear trend, as in Arellano (2008) and Chatterjee and Eyigungor (2012). This gives $\rho = 0.77$ and $\sigma_\varepsilon = 0.026$. The risk-free rate is $r = 1.8$ percent a year in real terms, following Bocola et al. (2019). The reentry probability is $\theta = 0.25$, a mean exclusion of four years. Colombia offers no default from which to estimate it, and estimates of the length of exclusion after a default range from about three to six years, depending on the definition of market access and on the size of the haircut (Gelos et al., 2011; Cruces and Trebesch, 2013). Appendix A.16 reports the calibration under other values of θ. Coupons are normalized to zero, so each bond is a geometric-decay perpetuity whose Macaulay duration is $D = (1+r)/(\lambda+r)$. We set each λ so that this duration equals the median duration at issuance of Colombia's debt under that law (Section 2.3): 3.75 years for domestic law and 8.71 years for foreign law, which gives $\lambda_D = 0.254$ and $\lambda_F = 0.099$.

We choose the remaining five parameters jointly to match five moments of Colombia's data over 2006–2024: the discount factor β, the two recoveries ν_D and ν_F, and the two

parameters of the output loss in default, which follows Chatterjee and Eyigungor (2012),

$$\phi(y) = \max\left\{0,\, d_0\, y + d_1\, y^2\right\}.$$

Three of them are the moments that quantitative models of sovereign default usually target (Arellano, 2008; Hatchondo and Martinez, 2009; Chatterjee and Eyigungor, 2012; Arellano and Ramanarayanan, 2012): the mean and the volatility of the spread, 2.40 and 0.75 percent, from the J.P. Morgan EMBI Global Colombia, and the ratio of marketable debt to GDP, 20.9 percent (Section 2.2). The other two are new and discipline the recoveries: the foreign-law share of marketable debt, 29 percent (Section 2.2), and the slope of the yield gap on the CDS spread (Section 2.4). Nearly all of Colombia's foreign-law debt is issued under New York law, 98 percent of the face value it issued under foreign law (Table 13), and the EMBI spread, the yield gap and the issuance fee of Section 2.5 all come from New York-law bonds, so we read the foreign-law bond in the model as a New York-law bond and its spread as the model counterpart of the EMBI spread. Debt in the model is the face value of both bonds over output.

We target how the yield gap moves with default risk and leave its average level untargeted. The average difference between the yields of the two laws contains the value of governing law, but also every other difference between the two bonds that does not depend on default risk: their liquidity, the investors who hold them, their tax treatment, their inclusion in bond indices, and the error in fitting the domestic-law curve. The value of a higher recovery is different, because investors collect it only if the government defaults. It is worth little when default is unlikely and rises with the probability of default. The slope of the gap on the CDS spread isolates this component, and differencing removes the wedges that stay constant. Targeting the level would attribute all of these wedges to the recoveries. In the model, the gap is zero whenever $\nu_D = \nu_F$, and it rises with the probability of default at a rate that increases with $\nu_F - \nu_D$.

To compute the slope in the model, we price bonds that resemble those in the data. In the data, the gap is measured on a panel of bonds whose remaining maturity shortens every day, and the regression in Section 2.4 absorbs this with time to maturity and bond fixed effects. The model has one bond under each law, each with a constant duration. Along the simulated equilibrium path, using the solved default and borrowing policies, we therefore price synthetic bonds with constant average maturities $M \in \{1, 3, 5, 7, 10\}$ years and decay $\lambda_M = 1/M$. At each maturity we price a domestic-law and a foreign-law version, which differ only in their recovery. The yield gap at maturity M is the difference

between their spreads,

$$s^D_{M,t} - s^F_{M,t} = \lambda_M \left(\frac{1}{q^D_{M,t}} - \frac{1}{q^F_{M,t}} \right), \tag{16}$$

and the model counterpart of the CDS spread is the spread on the five-year foreign-law bond, $s^F_{5,t}$. The model slope γ is the estimate of

$$\Delta\left(s^D_{M,t} - s^F_{M,t}\right) = \gamma\, \Delta s^F_{5,t} + \alpha_M + \varepsilon_{M,t} \tag{17}$$

over the periods with market access. The synthetic bonds span the range of maturities in the data, the maturity fixed effect α_M plays the role of the bond fixed effect, and holding each maturity constant plays the role of the maturity control. The regression therefore uses the same variation as in the data: changes over time in the gap of a given bond. The model is real, so its slope has no currency effect; we target the slope in the data, and Appendix A.15 reports the calibration with the slope corrected for the fall in the peso at default.

Table 7. Parameters and Targets

A. Externally set				
Parameter	Value	Description	Source	
σ	2	Relative risk aversion	Standard	
ρ	0.77	Autocorrelation of log output	WDI, 1976–2024	
σ_ε	0.026	Std. deviation of output shocks	WDI, 1976–2024	
r	0.018	Risk-free rate	Bocola et al. (2019)	
θ	0.25	Reentry probability	Four-year exclusion	
λ_D	0.254	Decay, domestic-law debt	Duration, 3.75 years	
λ_F	0.099	Decay, foreign-law debt	Duration, 8.71 years	
B. Internally calibrated				
Parameter	Value	Main target	Data	Model
β	0.920	Marketable debt / GDP (%)	20.9	21.0
d_0	-0.188	Mean spread (%)	2.40	2.34
d_1	0.236	Spread volatility (%)	0.75	1.02
ν_D	0.40	Foreign-law share (%)	29.0	29.0
ν_F	0.469	Slope of the yield gap, γ	0.23	0.23

Notes: Annual frequency. Colombia, 2006–2024. The parameters in panel B are chosen jointly to match the five moments; each is listed next to the moment it affects most. The spread is the J.P. Morgan EMBI Global Colombia spread, and its model counterpart is the spread on the foreign-law bond.

Table 7 pairs each parameter with the moment it affects most, although all five are chosen

jointly. The mean spread mainly disciplines d_0, its volatility d_1, and the debt ratio β. The two recoveries are pinned down jointly by the two new moments, as Figure 4 shows. The slope of the yield gap rises with the recovery gap $\nu_F - \nu_D$ (panel a). Holding that gap fixed, the foreign-law share rises with the level of recovery (panel b): as both recoveries rise, the government borrows more and defaults more often, so the same recovery gap applies in more states. The calibration delivers a recovery of 40 percent on domestic-law debt and 47 percent on foreign-law debt.

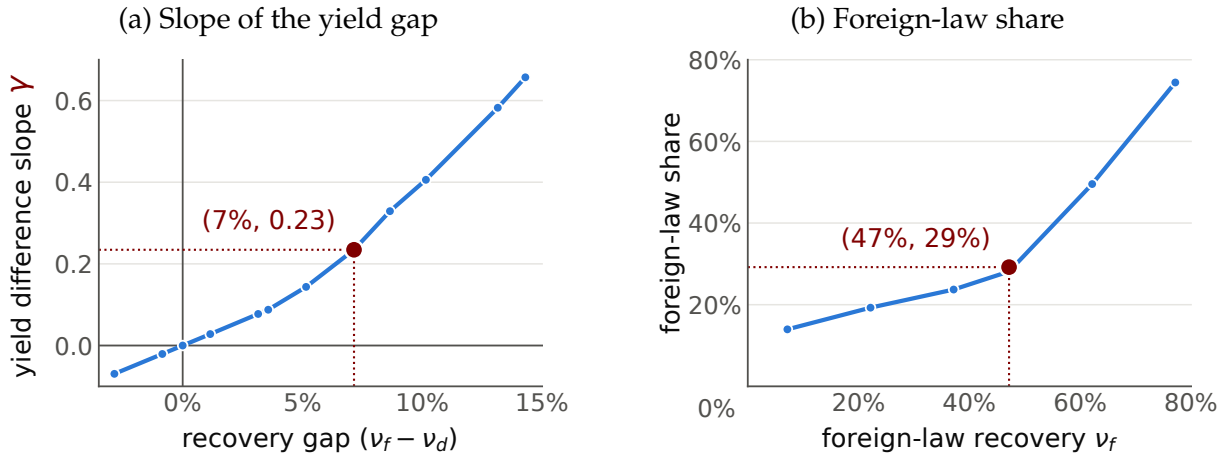


Figure 4. Identification of the Recoveries

Notes: Panel (a) plots the model slope of the yield gap on the CDS spread, γ, against the recovery gap $\nu_F - \nu_D$. Panel (b) plots the foreign-law share of debt against the foreign-law recovery ν_F, holding the recovery gap at its calibrated value. Other parameters at their calibrated values. The dots mark the calibration.

4.2 Model and data

As Table 7 shows, the model closely matches the targeted moments, although it overstates the volatility of the spread, 1.02 percent against 0.75 percent in the data.

Table 8 compares the model with moments we do not target. The model reproduces the volatility of consumption relative to output, 1.17 against 1.15 in the data, and the burden of debt service, 4.3 percent of GDP against 3.9 percent, although debt service in the model excludes coupons. Spreads and the foreign-law share of new issuance are more countercyclical in the model than in the data: their correlations with output are -0.69 in the model, against -0.12 and -0.29 in the data. As in other models in which output is the only shock (Arellano, 2008), default risk in the model rises only when output falls, so spreads move more closely with output than they do in the data. The model implies that Colombia defaults in 3.7 percent of years, a frequency that, like the recoveries, follows from Colombia's own prices and debt.

Table 8. Untargeted Moments

	Model	Data
corr(spread, y)	-0.69	-0.12
corr(foreign-law share of issuance, y)	-0.69	-0.29
$\sigma(c)/\sigma(y)$	1.17	1.15
corr(c, y)	0.83	0.92
Debt service / GDP (%)	4.3	3.9
$\sigma(\log y)$	0.039	0.041
Default frequency (%)	3.7	—

Notes: Colombia, 2006–2024. Data moments for output and consumption use the cyclical component of logged series, HP filter with $\lambda = 100$, except $\sigma(\log y)$, which is implied by the linearly detrended AR(1); model moments use the simulated series. Model debt service includes amortization only, because coupons are zero; the data include coupons.

The model also generates the split between the two laws that we observe: the government holds 29 percent of its debt under foreign law and the rest under domestic law. Section 4.3 shows that this split requires the two bonds to differ in both recovery and duration.

4.3 Experiments: the dilution–default trade-off

The government in the calibrated model borrows under both laws because each law addresses a different commitment problem. To show this, we run two experiments. In each, we remove one of the two differences between the bonds, keep every other parameter at its calibrated value, and solve the model again. In both, the government ends up using a single bond. Appendix A.17 reports the full results.

When the two bonds have the same duration, the government issues only the bond with the higher recovery. We give both bonds a common duration, on a grid from one to 8.71 years, and vary the two recoveries. At every duration, the government issues only the bond with the higher recovery, even when the recoveries differ by 0.02, and reversing which law recovers more reverses the result (Table 25). A higher recovery lowers the loss investors bear in default, so it lowers the compensation for default risk that the government pays on every unit of debt. Lemma 1 shows that this ranking of prices holds in every equilibrium of the model, whatever the government's borrowing and default policies. With equal durations, nothing offsets this advantage.

Lemma 1. *Suppose $\lambda_D = \lambda_F$ and $\nu_F \geq \nu_D$. Then, in any equilibrium and in every state,*

$$q_F^{ND}(y, b_D, b_F) \geq q_D^{ND}(y, b_D, b_F) \quad \text{and} \quad q_F^{D}(y, b_D, b_F) \geq q_D^{D}(y, b_D, b_F).$$

If $\nu_F > \nu_D$, the inequalities are strict in every state from which, with positive probability, the government later reenters and then repays.

Proof. Appendix A.18. □

When the two bonds have the same recovery, the government issues almost only the shorter bond. We give both bonds a common recovery, on a grid from 0 to 0.75, and keep the calibrated durations of 3.75 years for domestic law and 8.71 years for foreign law. The foreign-law share is between 0.1 and 0.3 percent at every recovery (Table 27). The government cannot commit to its future borrowing, and new borrowing lowers the price of debt still outstanding. Long debt remains outstanding longer, so it is more exposed to this dilution, and investors pay less for it. The long bond keeps a role only when the short bond is much shorter: with a one-year domestic-law bond, the government holds between 48 and 68 percent of its debt in the long bond.

Each law therefore protects investors against one of the two risks. Foreign-law debt recovers more, so it protects them against default. Domestic-law debt is shorter, so it protects them against dilution. When the bonds differ in only one of these dimensions, the government uses only one bond. The split we observe arises because the bond with the higher recovery is also the longer bond, and Section 5.1 asks why the government does not issue a short bond with the higher recovery.

5 The Fee on Foreign-Law Debt

5.1 Why governments do not issue short foreign-law bonds

We now use the model to ask why Colombia does not issue a short foreign-law bond, which would protect investors against both default and dilution: it would carry the higher recovery of foreign law and the short duration of domestic-law debt. Between 2006 and 2024, Colombia's shortest foreign-law bond had a duration of 7.2 years at issuance, while its shortest domestic-law bond had a duration of 0.2 years. The fee documented in Section 2.5 is enough to explain this, but only because the government can offer investors the recovery of foreign law through a long bond. We add the fee to the model of

Section 4 and run two experiments. In the first, the government chooses between a one-period domestic-law bond and a one-period foreign-law bond. In the second, we add a long foreign-law bond.

The government pays a fee f per unit of face value it issues under foreign law, and only when it issues. With foreign-law bonds collected in the set $\mathcal{F}$, the budget constraint becomes

$$c + \sum_j \lambda_j b_j \;=\; y + \sum_j q_j \left[b'_j - (1-\lambda_j) b_j \right] \;-\; f \sum_{j \in \mathcal{F}} \max\left\{0,\; b'_j - (1-\lambda_j) b_j \right\}. \tag{18}$$

A buyback pays no fee and earns no rebate. As in the data, the fee is a share of the face value issued, and the share does not depend on the maturity of the bond or on the amount issued. For Colombia we set $f = 0.0025$, the median fee of 25 basis points on its New York-law bonds (Table 5). Colombia's fee does not depend on the maturity or the size of its bonds either. In the same regression on its 49 New York-law bonds, neither coefficient is significant, and since 2020 all 17 of its New York-law bonds have paid 20 basis points, with maturities from 10 to 40 years and sizes from US$0.3 to 2.2 billion (Appendix A.12). The pricing equations are those of Section 3, one for each bond, so the fee affects prices only through the government's default and borrowing policies.

The fee is a tax on rolling over debt. To keep one unit of face outstanding, the government issues each period the share λ that matures, and pays $f\lambda$ in fees. A one-period bond pays the fee on its whole face every period, while a long bond pays it on a fraction of its face. In terms of yields, the fee raises the cost of borrowing by approximately $\lambda f / q^2$, which is larger on the shorter bond (Appendix A.19).

With only one-period bonds, the fee does little. As in Section 4.3, without a fee the government borrows only under foreign law, because the two bonds are equally short and foreign-law debt recovers more. At 20 basis points it still holds 75 percent of its debt under foreign law, and it stops issuing under foreign law only at a fee of about 76 basis points. With only short bonds available, the only way to avoid the fee is to give up the higher recovery, and the government does so only when the fee is several times the one we measure.

The fee needed to keep the government from issuing short foreign-law bonds rises with default risk. The fee is a fixed charge per unit of face value, while the value of the higher recovery under foreign law rises with the probability of default. In the economy with three bonds, when the probability of default in the next year is below 0.5 percent, a fee of

about 20 basis points removes short foreign-law issuance. When the probability exceeds 2 percent, removing it takes about 45 basis points. At a fee of 20 basis points, short foreign-law bonds are 0.1 percent of the debt issued when the probability of default is below 0.5 percent, and 9 percent when it exceeds 1 percent. The short foreign-law borrowing that a fee leaves is therefore borrowing done when default is most likely.

Adding a long foreign-law bond changes the result. In the second experiment, the government chooses among a short domestic-law bond, a short foreign-law bond and a long foreign-law bond. The two foreign-law bonds share the recovery ν_F, and the domestic-law bond recovers ν_D. The short bonds last one period, $\lambda_D = \lambda_{Fs} = 1$, and the long bond has a duration of 2.32 years, $\lambda_{Fl} = (1+r)/2.32 - r = 0.420$, so that the ratio of the two durations matches Colombia's, $8.71/3.75 = 2.32$. These durations are shorter than in the data: we use one-period short bonds to keep the computation tractable, and we choose the long bond so that the ratio of the two durations matches Colombia's.[1] All other parameters keep their values from Section 4.1. Without a fee, the government holds short foreign-law debt worth 6.7 percent of GDP. At Colombia's fee of 25 basis points, this falls to 0.5 percent of GDP, a reduction of more than 90 percent, while total debt stays at 19 percent of GDP. Short foreign-law debt disappears at a fee of about 35 basis points, less than half the fee needed with only one-period bonds. A flat cost per issue of the size reported in the prospectuses, about US$300,000, leaves these results unchanged.

The difference between the two experiments is the long bond. It carries the same recovery as the short foreign-law bond, so the government can offer investors the protection of foreign law through it while paying the fee less often. Avoiding the fee then no longer means giving up the higher recovery: the government only moves its foreign-law borrowing to the long bond, and a fee the size of the one we measure is enough for it to do so. Its short borrowing moves home, to the bond that protects investors against dilution but offers them less protection in default. Section 5.2 measures what Colombia loses because that bond recovers less than foreign-law debt.

5.2 What the fee costs

We now use the model to measure what the fee costs Colombia. Without the fee, the government would replace its short domestic-law bond with a short foreign-law bond, which has the same duration and a higher recovery. In the model, this is achieved by raising the recovery of the short bond from 40 to 47 percent, keeping every other parameter at its

[1] With three bonds the state has four variables: output and three debt stocks. We solve the model on a grid of 25 points for output and 60 for each stock, $25 \times 60^3 \approx 5.4$ million states.

calibrated value. This raises welfare by 8.2 basis points of permanent consumption. For comparison, eliminating aggregate risk and the costs of default in the baseline economy raises welfare by 71 basis points, so removing the fee delivers about 12 percent of that gain.

We measure welfare as the permanent percentage change in consumption Δ that makes the household indifferent between the baseline economy $\mathcal{E}_0$ and the economy with higher domestic-law recovery $\mathcal{E}_1$,

$$(1+\Delta)^{1-\sigma}\,\bar{V}^{\mathcal{E}_0} = \bar{V}^{\mathcal{E}_1}, \qquad \bar{V}^{\mathcal{E}} \equiv \sum_{y} \pi(y)\, V^{\mathcal{E}}(y,b), \tag{19}$$

where π is the invariant distribution of output and b is the debt at which we evaluate the gain. We evaluate it at two levels of debt: at zero debt, as in Chatterjee and Eyigungor (2012) and Hatchondo et al. (2016), and at the mean level of debt in the baseline economy. In the new economy we solve the model again, so the government's borrowing and default policies adjust to the higher recovery. At zero debt the gain is 8.2 basis points; at the mean level of debt it is 9.0.

Table 9. Raising the Recovery of Domestic-Law Debt from 0.40 to 0.47

A. Change in the economy	
Mean spread, domestic-law debt (bp)	−46
Mean spread, foreign-law debt (bp)	−22
Debt / GDP (percentage points)	+0.4
Default frequency (percentage points a year)	−0.3
B. Welfare gain (bp of permanent consumption)	
Total	+8.2
New prices, policies fixed	+2.1
New debt at reentry, policies fixed	−1.3
Both, policies fixed	+0.8
Change in borrowing and default policies	+7.4

Notes: Changes from the baseline economy, $\nu_D = 0.40$, to the economy with $\nu_D = 0.47$, model simulations. The welfare gain is Δ in equation (19), evaluated at zero debt. The change in policies is the total minus the gain with both changes and fixed policies. Appendix A.22 describes the decomposition.

Table 9 shows how the economy changes. With the same recovery under both laws, the government borrows more and defaults less often: debt rises by 0.4 percentage points

of GDP, and the default frequency falls by 0.3 percentage points a year. Spreads fall by 46 basis points on domestic-law debt and by 22 basis points on foreign-law debt. In the baseline, the government offers investors protection against default by borrowing long under foreign law, which exposes that debt to more dilution. With the same recovery on its short domestic-law bond, it can offer investors that protection without lengthening its debt, and it moves its borrowing to the short bond.

A higher recovery affects welfare through three forces. First, investors expect to recover more in default, so they pay more for the government's debt. Second, the government comes out of default owing more. Third, the government changes how much it borrows and when it defaults. To measure the first two, we keep the government's borrowing and default policies at their baseline values and change one feature at a time. Giving the government the prices of the new economy, with the baseline debt at reentry, raises welfare by 2.1 basis points. Giving it the larger debt at reentry of the new economy, with baseline prices, lowers welfare by 1.3 basis points. With both changes and fixed policies, welfare rises by 0.8 basis points, so the remaining 7.4 basis points of the total come from the government's response to the higher recovery. The literature on governing law emphasizes that stronger legal protection lowers a government's borrowing costs when default risk is high (Chamon et al., 2018). In our model this is the smaller part of the gain: most of it comes from a shift in the government's incentives to borrow and to default. Appendix A.22 describes these calculations.

Part of the gain comes from letting an impatient government borrow more. As is standard in quantitative models of sovereign default, the calibrated government discounts the future more than investors do, and this impatience may reflect political frictions more than the preferences of households (Aguiar et al., 2020). We check that the gain does not depend on this impatience. We evaluate it with a household discount factor β_H that ranges from that of investors, $1/(1+r) = 0.982$, to that of the government, $\beta = 0.92$, holding the government's policies and the prices they imply fixed.

To show where the gain comes from, we follow Aguiar et al. (2020) and split the value of market access into three terms. The default-cost term is the welfare lost because output falls while the government is excluded; we obtain it by comparing the economy with one in which exclusion carries no output loss, but the government still comes out of default owing only the recovered share of its debt. The front-loading term is the value of bringing consumption forward by borrowing; we obtain it by comparing a household that receives for sure the mean consumption path of the economy with one that consumes mean output forever. An impatient household values front-loading, and a patient one does not. The

variability term measures how much more or less volatile consumption is with market access than in autarky. The value of market access is the product of these three terms, and the welfare gain is the ratio of its values in the two economies, so each term contributes the ratio of its own values. Appendix A.22 gives the formulas.

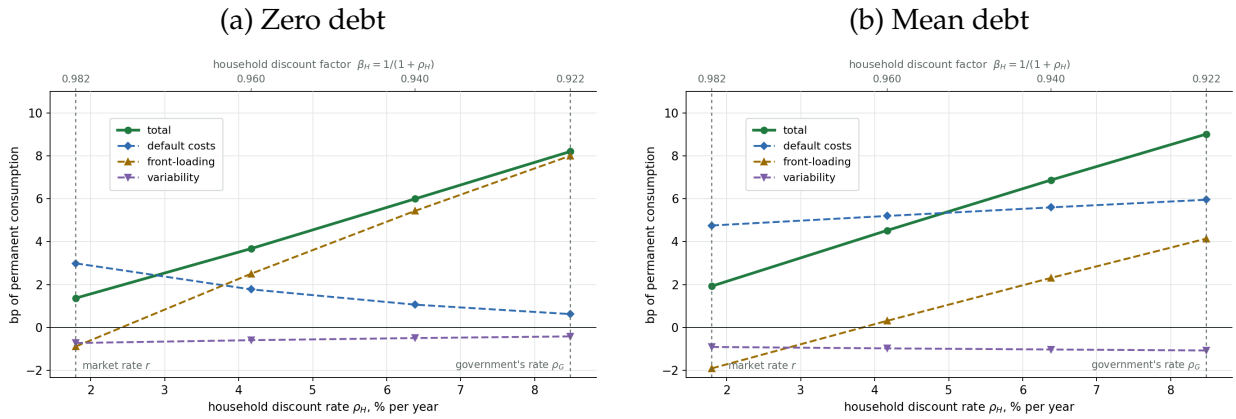


Figure 5. Welfare Gain and the Household's Discount Rate

Notes: Welfare gain from raising ν_D from 0.40 to 0.47, in basis points of permanent consumption, against the household's discount factor, from that of investors, $1/(1+r)$, to that of the government, β. The total is the product of the default-cost, variability and front-loading terms defined in Appendix A.22. Panel (a) evaluates the gain at zero debt and panel (b) at the mean level of debt in the baseline economy.

Figure 5 plots the gain and its three terms against the household's discount factor, at zero debt (panel a) and at the mean level of debt (panel b). Households gain at every discount factor between that of investors and that of the government. At zero debt, the total gain rises from 1.4 basis points for a household as patient as investors to 8.2 basis points for one as impatient as the government. The two ends differ in what drives the gain. A patient household values the lower costs of default, worth 3.0 basis points, while the additional borrowing lowers its welfare: its front-loading term is -0.9 basis points. As the household becomes more impatient, it values the additional borrowing more; the front-loading line rises until it accounts for almost all of the gain at the government's discount factor, while the default-cost line falls to 0.6 basis points. The variability term is small and slightly negative throughout, because the government borrows more and consumption becomes slightly more volatile. At the mean level of debt, the lower costs of default matter more: they contribute between 4.7 and 6.0 basis points at every discount factor, and the total gain ranges from 1.9 to 9.0 basis points.

6 Conclusion

Emerging-market governments borrow both under their own law and under foreign law, usually that of New York or England. If the government defaults, the governing law determines whether domestic or foreign courts settle disputes, and so how much investors recover. This paper asks why governments use both laws, and how much investors expect to recover under each law when the government defaults. We answer in three steps.

First, foreign law becomes more valuable to investors as default risk rises. We compare bonds that pay in the same currency under each law in Brazil, Colombia, Paraguay and Russia. In all four, the yield gap between the two laws rises with the probability of default: as default becomes more likely, investors demand a higher yield on domestic-law bonds.

Second, each law protects investors against a different risk. In a quantitative model of sovereign default calibrated to Colombia, the share of debt the government holds under each law, together with how the yield gap moves with default risk, pins down how much investors expect to recover. They expect to recover 47 percent of what they are owed under foreign law and 40 percent under domestic law. Foreign-law debt therefore protects them in default, and domestic-law debt, which is shorter, protects them against the government's future borrowing.

Third, the fee that banks charge to place foreign-law bonds explains why governments do not issue short foreign-law bonds. The fee is a share of face value, so a short bond pays it more often. A fee of the size we measure is enough to keep the government from issuing them. Removing the fee would raise welfare by 8.2 basis points of permanent consumption, about 12 percent of the gain from eliminating aggregate risk and default costs.

References

Aguiar, Mark and Gita Gopinath, "Defaultable debt, interest rates and the current account," *Journal of International Economics*, 2006, *69* (1), 64–83. Emerging Markets.

_ , Manuel Amador, and Stelios Fourakis, "On the Welfare Losses from External Sovereign Borrowing," *IMF Economic Review*, 2020, *68*, 163–194.

_ , _ , Hugo Hopenhayn, and Iván Werning, "Take the short route: Equilibrium default and debt maturity," *Econometrica*, 2019, *87* (2), 423–462.

Altınkılıç, Oya and Robert S. Hansen, "Are There Economies of Scale in Underwriting Fees? Evidence of Rising External Financing Costs," *Review of Financial Studies*, 2000, *13* (1), 191–218.

Arellano, Cristina, "Default risk and income fluctuations in emerging economies," *American economic review*, 2008, *98* (3), 690–712.

_ and Ananth Ramanarayanan, "Default and the maturity structure in sovereign bonds," *Journal of Political Economy*, 2012, *120* (2), 187–232.

_ and Leonardo Barreto, "Official Sovereign Debt," Working Paper, Federal Reserve Bank of Minneapolis 2025.

_ , Xavier Mateos-Planas, and José-Víctor Ríos-Rull, "Partial Default," *Journal of Political Economy*, 2023, *131* (6), 1385–1439.

Asonuma, Tamon and Christoph Trebesch, "Sovereign Debt Restructurings: Preemptive or Post-Default," *Journal of the European Economic Association*, 2016, *14* (1), 175–214.

Benjamin, David and Mark L. J. Wright, "Recovery Before Redemption: A Theory of Delays in Sovereign Debt Renegotiations," CAMA Working Paper 15/2009, Centre for Applied Macroeconomic Analysis, Australian National University 2009.

Bigio, Saki, Galo Nuño, and Juan Passadore, "Debt-Maturity Management with Liquidity Costs," 2022. Working paper, October 2022.

Bocola, Luigi, Gideon Bornstein, and Alessandro Dovis, "Quantitative sovereign default models and the European debt crisis," *Journal of International Economics*, 2019, *118*, 20–30.

Broner, Fernando A., Guido Lorenzoni, and Sergio L. Schmukler, "Why Do Emerging Economies Borrow Short Term?," *Journal of the European Economic Association*, 2013, *11* (S1), 67–100.

Chamon, Marcos, Julian Schumacher, and Christoph Trebesch, "Foreign-law bonds: Can they reduce sovereign borrowing costs?," *Journal of International Economics*, 2018, *114*, 164–179.

Chari, V. V., Alessandro Dovis, and Patrick J. Kehoe, "On the Optimality of Financial Repression," *Journal of Political Economy*, 2020, *128* (2), 710–739.

Chatterjee, Satyajit and Burcu Eyigungor, "Maturity, indebtedness, and default risk," *American Economic Review*, 2012, *102* (6), 2674–2699.

_ and _, "A seniority arrangement for sovereign debt," *American Economic Review*, 2015, *105* (12), 3740–3765.

Choi, Jaewon, Sebastiao Oliveira, Jay Rafi, and Josef Zechner, "Optimal Sovereign Debt Structure: Evidence and Theory," 2026. Working paper, September 28, 2026.

Cruces, Juan J and Christoph Trebesch, "Sovereign defaults: The price of haircuts," *American economic Journal: macroeconomics*, 2013, *5* (3), 85–117.

Deng, Mingzhuo, Pablo A. Guerrón-Quintana, and Lewis Tseng, "Parallel Computation of Sovereign Default Models," *Computational Economics*, 2023, *62*, 1047–1085.

Du, Wenxin and Jesse Schreger, "Local currency sovereign risk," *The Journal of Finance*, 2016, *71* (3), 1027–1070.

_ , Carolin E. Pflueger, and Jesse Schreger, "Sovereign Debt Portfolios, Bond Risks, and the Credibility of Monetary Policy," *The Journal of Finance*, 2020, *75* (6), 3097–3138.

Eaton, Jonathan and Mark Gersovitz, "Debt with Potential Repudiation: Theoretical and Empirical Analysis," *The Review of Economic Studies*, 1981, *48* (2), 289–309.

Erce, Aitor, Enrico Mallucci, and Mattia Osvaldo Picarelli, "Sovereign defaults at home and abroad," 2024.

_ , _ , and Mattia Picarelli, "A Journey in the History of Sovereign Defaults on Domestic-Law Public Debt," International Finance Discussion Papers 1338, Board of Governors of the Federal Reserve System 2022.

Fang, Chuck, Julian Schumacher, and Christoph Trebesch, "Restructuring Sovereign Bonds: Holdouts, Haircuts and the Effectiveness of CACs," *IMF Economic Review*, 2021, *69*, 155–196.

Flandreau, Marc, Juan H. Flores, Norbert Gaillard, and Sebastián Nieto-Parra, "The End of Gatekeeping: Underwriters and the Quality of Sovereign Bond Markets, 1815–

2007," in "NBER International Seminar on Macroeconomics 2009," National Bureau of Economic Research, 2010, pp. 53–92.

Gelos, R. Gaston, Ratna Sahay, and Guido Sandleris, "Sovereign Borrowing by Developing Countries: What Determines Market Access?," *Journal of International Economics*, 2011, *83* (2), 243–254.

Gordon and Guerron-Quintana, "Dynamics of investment, debt, and default," *Review of Economic Dynamics*, 2018, *28*, 71–95.

Hatchondo, Juan Carlos and Leonardo Martinez, "Long-duration bonds and sovereign defaults," *Journal of International Economics*, 2009, *79* (1), 117–125.

_ , _ , and Cesar Sosa-Padilla, "Debt dilution and sovereign default risk," *Journal of Political Economy*, 2016, *124* (5), 1383–1422.

Jeanne, Olivier, "Debt Maturity and the International Financial Architecture," *American Economic Review*, 2009, *99* (5), 2135–2148.

Jungherr, Joachim, Matthias Meier, Timo Reinelt, and Immo Schott, "Corporate Debt Maturity Matters for Monetary Policy," International Finance Discussion Papers 1402, Board of Governors of the Federal Reserve System 2024.

Lee, Inmoo, Scott Lochhead, Jay Ritter, and Quanshui Zhao, "The Costs of Raising Capital," *Journal of Financial Research*, 1996, *19* (1), 59–74.

Macaulay, Frederick R., *Some Theoretical Problems Suggested by the Movements of Interest Rates, Bond Yields and Stock Prices in the United States since 1856*, New York: National Bureau of Economic Research, 1938.

Mano, Rui C., "Exchange Rates upon Sovereign Default," 2013. Job market paper, University of Chicago.

Melnik, Arie and Doron Nissim, "Debt Issue Costs and Issue Characteristics in the Market for U.S. Dollar Denominated International Bonds," *European Finance Review*, 2003, *7* (2), 277–296.

Mihalache, Gabriel, "Sovereign default resolution through maturity extension," *Journal of International Economics*, 2020, *125*, 103326.

Missale, Alessandro and Olivier Jean Blanchard, "The Debt Burden and Debt Maturity," *American Economic Review*, 1994, *84* (1), 309–319.

Na, Seunghoon, Stephanie Schmitt-Grohé, Martín Uribe, and Vivian Yue, "The Twin Ds: Optimal Default and Devaluation," *American Economic Review*, 2018, *108* (7), 1773–1819.

Ottonello, Pablo and Diego J. Perez, "The Currency Composition of Sovereign Debt," *American Economic Journal: Macroeconomics*, 2019, *11* (3), 174–208.

Passadore, Juan and Yu Xu, "Illiquidity in sovereign debt markets," *Journal of International Economics*, 2022, *137*, 103618.

_ , Francisco Roldán, and Yu Xu, "The Term Structure of Credit and Liquidity Risk Premia," 2026. Work in progress.

Reinhart, Carmen M., "Default, Currency Crises, and Sovereign Credit Ratings," *The World Bank Economic Review*, 2002, *16* (2), 151–170.

_ and M. Belen Sbrancia, "The Liquidation of Government Debt," *Economic Policy*, 2015, *30* (82), 291–333.

Saiegh, Sebastian M. and Glen Biglaiser, "The Value of Legal Recourse in Sovereign Bond Markets: Evidence from Argentina," *Journal of Empirical Legal Studies*, 2024, *21*, 669–709.

Schumacher, Julian, Christoph Trebesch, and Henrik Enderlein, "Sovereign Defaults in Court," Kiel Working Paper 2103, Kiel Institute for the World Economy 2018.

Sturzenegger, Federico and Jeromin Zettelmeyer, "Haircuts: Estimating Investor Losses in Sovereign Debt Restructurings, 1998–2005," *Journal of International Money and Finance*, 2008, *27*, 780–805.

von Luckner, Clemens M Graf, Josefin Meyer, Carmen M Reinhart, and Christoph Trebesch, "Sovereign haircuts: 200 years of creditor losses," *IMF Economic Review*, 2025, *73* (1), 150–195.

Yue, Vivian Z., "Sovereign Default and Debt Renegotiation," *Journal of International Economics*, 2010, *80* (2), 176–187.

Zettelmeyer, Jeromin, Christoph Trebesch, and Mitu Gulati, "The Greek debt restructuring: an autopsy," *Economic Policy*, 2013, *28* (75), 513–563.

A Appendix

A.1 Shares of outstanding debt

Inflation-indexed bonds enter the domestic-law stock at their accreted capital: the indexed amount times the current value of the index, which is what the government owes. We convert the domestic-law stock to dollars at the annual average of the year's daily exchange rates. GDP is converted at the same rate, so the exchange rate does not affect debt as a percentage of GDP. The foreign-law share does not involve GDP, so for it the 31 December rate would match the date at which both stocks are measured; Table 10 reports that alternative. A foreign-law tranche counts as foreign law unless its governing-law field in Dealogic names the home country, so a tranche with a blank field falls on the foreign side.

Table 10. Foreign-Law Share of Marketable Debt: Robustness

	Headline (1)	Dec. FX (2)	Med. FX (3)	Revalued (4)	Net pub. (5)
Brazil	5.0	5.2	5.0	5.1	—
Chile	23.8	24.2	23.7	23.7	23.8
Colombia	29.7	30.2	29.4	29.8	31.6
Hungary	30.2	30.5	30.0	30.2	30.6
Mexico	16.0	16.3	15.9	15.8	—
Paraguay	51.9	52.2	51.9	51.9	—
Peru	34.7	34.8	34.8	34.9	35.5
Poland	20.8	20.8	20.7	20.6	22.2
Russia	20.9	21.6	20.8	20.8	23.5
South Africa	8.5	8.6	8.5	8.7	8.5
Turkey	24.5	26.3	24.5	24.4	28.0
Uruguay	62.5	62.9	62.6	62.4	—

Notes: Column (1) repeats column (4) of Table 1. Columns (2) and (3) convert domestic-law stocks at the 31 December exchange rate and at the annual median of daily rates, instead of the annual average. Column (4) revalues foreign-law bonds at each year's exchange rate instead of keeping their dollar value at issue. Column (5) also removes holdings of public institutions. — : not available.

Table 10 reports four alternatives to the baseline share. Columns (2) and (3) convert the domestic-law stock at the 31 December exchange rate and at the annual median of daily rates. Column (4) revalues each foreign-law bond at each year's exchange rate, using the dual-currency face values that Dealogic records, instead of keeping its dollar value on the issue date. Column (5) also removes the domestic-law holdings of public institutions

that bought at competitive auctions. A last exercise, not in the table, removes from the foreign-law side only the tranches with a blank governing-law field whose law cannot be established from another source.

A.2 Duration differences

Each curve in Figure 1 is a Gaussian kernel density estimated separately for one country and one law, over unweighted placements. The bandwidth is 0.45 times the standard deviation of that series, so that the curves of the two laws are smoothed by the same amount relative to their own dispersion, although domestic-law series have far more placements. No correction is applied at zero, so for countries whose domestic-law issuance is dominated by short bills the curve understates the mass near zero; the medians and the statistics in the tables are computed from the placements and are not affected.

For bills and zero-coupon bonds, duration equals maturity. For fixed-coupon bonds we apply the formula of Section 2.3 to their coupons and principal, and for inflation-linked bonds we apply it in real terms, with the real coupon and the real yield. For floating-rate bonds, duration equals the time to the next coupon reset. The yield at issuance comes from official sources for domestic-law placements and from Dealogic for foreign-law placements. Duration for a floating-rate bond depends on how we treat its coupon resets, so maturity, which counts every bond over its full term, gives a measure free of that choice; Table 11 reports it, summarised as in Table 2. Table 12 recomputes the medians with every country on 2006–2024, and Table 13 splits foreign-law face by the law under which each placement is written.

Table 11. Maturity at Issuance by Governing Law

	Domestic law				Foreign law				Gap	
	Median (1)	Mean (2)	Face (3)	GDP (4)	Median (5)	Mean (6)	Face (7)	GDP (8)	Median (9)	Face (10)
Brazil	4.59	7.29	4.45	4.36	10.40	16.41	16.24	16.46	5.81	11.79
Chile	9.84	13.81	10.67	10.81	11.42	16.48	16.83	16.60	1.58	6.16
Colombia	4.08	6.64	8.41	—	11.00	17.79	17.76	—	6.92	9.35
Hungary	3.85	4.94	3.08	2.91	8.50	9.39	10.82	10.64	4.65	7.74
Mexico	1.00	4.98	3.31	3.23	10.00	17.25	19.74	19.30	9.00	16.43
Paraguay	3.83	4.92	5.28	4.96	11.41	16.79	16.89	16.94	7.58	11.61
Peru	11.54	15.28	14.05	14.65	16.00	24.20	22.74	22.60	4.46	8.69
Poland	5.26	6.32	5.36	5.15	10.00	11.00	11.34	11.22	4.74	5.98
Russia	7.44	8.33	8.91	8.72	10.00	12.90	14.35	14.31	2.56	5.44
South Africa	9.62	11.34	6.72	6.84	12.00	16.91	17.18	16.90	2.38	10.46
Turkey	2.99	3.89	3.54	3.16	10.00	10.95	10.94	11.37	7.01	7.40
Uruguay	9.53	9.90	10.14	10.00	12.00	16.20	18.92	19.01	2.47	8.78

Notes: Years from settlement to final payment of each placement. Columns as in Table 2. — : not available.

Table 12. Maturity and Duration at Issuance: Common Window

	Maturity		Duration	
	Domestic (1)	Foreign (2)	Domestic (3)	Foreign (4)
Brazil	4.59	10.40	3.47	8.40
Chile	9.84	11.42	8.85	9.90
Colombia	4.08	11.00	3.75	8.71
Hungary	4.29	9.00	2.94	7.43
Mexico	1.00	10.00	0.48	8.87
Paraguay	3.83	11.41	3.46	8.74
Peru	11.54	16.00	8.64	11.77
Poland	5.32	10.00	2.60	8.15
Russia	7.57	10.00	5.62	8.00
South Africa	9.58	12.00	6.89	9.06
Turkey	4.25	10.00	1.64	7.68
Uruguay	9.53	12.00	8.19	9.45

Notes: Median maturity and median duration at issuance, in years, with every country on 2006–2024.

Table 13. Governing Law of Foreign-Law Placements

Country	Governing law (% of foreign-law face)
Brazil	NY 84, London 9, Tokyo 4, Frankfurt 2, Lux 2
Chile	NY 100
Colombia	NY 98, Tokyo 2
Hungary	London 72, NY 21, Tokyo 4, China 2, Frankfurt 1
Mexico	NY 90, Tokyo 8, Frankfurt 2, London 1
Paraguay	NY 100
Peru	NY 100
Poland	London 66, NY 27, Tokyo 5, Frankfurt 1, China 1
Russia	London 96, Frankfurt 4
South Africa	NY 90, London 5, Tokyo 3, Frankfurt 2
Turkey	NY 64, London 23, Frankfurt 7, Tokyo 6
Uruguay	NY 93, Tokyo 7

Notes: Share of foreign-law face issued under each law, from most to least. NY includes tranches that name New York beside the home law.

A.3 A simple model of the yield gap

A government's bonds differ along two dimensions, the law that governs them and the currency they pay, and the two get confounded. Three configurations matter. Two bonds in the same hard currency, differing only in law, give the premium directly; they are almost never issued. A foreign-law bond in dollars against a domestic-law bond in local currency is the usual emerging-market pairing, and the currency conversion it requires fails when default and depreciation coincide. Two bonds in the same local currency, differing only in law, give the premium again, scaled by a factor below one.

Case 1: payoffs in the investor's currency. Suppose a government has two bonds outstanding. Both are zero-coupon bonds maturing next period, both are denominated in dollars, and they are identical except in the law that governs them, $j \in \{D, F\}$. An international investor prices both. She is risk-neutral, so she discounts every payoff at the gross risk-free dollar rate $1 + r^{\$}$. The government repays with probability $1 - p$ and defaults with probability p, and it defaults on the two instruments together. After a default a holder of the law-j bond recovers a fraction $\nu_j \in (0, 1)$ of face. Whether foreign law secures the larger recovery, $\nu_F > \nu_D$, is the hypothesis the empirical work tests.

The bond pays one dollar if the government repays and ν_j dollars if it defaults, so

$$q_j^{\$} = \frac{1}{1 + r^{\$}} \Big[(1 - p)\, 1^{\$} + p\, \nu_j\, 1^{\$} \Big], \tag{20}$$

where a superscript \$ marks a quantity denominated in dollars and a superscript LC, used from case 2, one denominated in local currency. Differencing the two prices cancels the common repayment payoff and leaves the recovery difference:

$$\Pi^{\$} \equiv q_F^{\$} - q_D^{\$} = \frac{p\,(\nu_F - \nu_D)\,1^{\$}}{1 + r^{\$}}. \tag{21}$$

The premium is the value of a recovery advantage $\nu_F - \nu_D$ collected only in default. Differentiating with respect to p,

$$\frac{\partial \Pi^{\$}}{\partial p} = \frac{(\nu_F - \nu_D)\,1^{\$}}{1 + r^{\$}}, \tag{22}$$

which is positive under $\nu_F > \nu_D$, so the premium widens with the default probability, and does not respond to it at all under the null $\nu_F = \nu_D$.

Prices are not comparable across bonds of different coupon and maturity, and for several countries in the sample more than one foreign-law bond is outstanding at a time. The premium is therefore measured as the gap between the domestic-law and foreign-law yields. The yield gap carries the sign of the price gap. Since both prices fall as default becomes more likely, it widens with default risk wherever the price gap is positive. The last paragraphs of this section set out the mapping.

The price premium rises with maturity. The yield premium need not: turning a price gap into a rate per year divides it by the bond's horizon and by its price, and lengthening maturity raises the first and lowers the second, so the direction depends on the parameters. The last paragraphs of this section derive both. Residual maturity therefore enters the specification as a control.

The model gives the two bonds the same trading costs, but in practice their liquidity differs (Chamon et al., 2018), so part of any measured gap may reflect differences in the cost of trading. To control for this, the specification includes the foreign-law bond's bid-ask spread relative to that of the country's domestic ten-year benchmark; the domestic-law bond with the same payments is a constructed object and has no quoted spread of its own.

Case 2: currency conversion with a forward. Now suppose instead that the two bonds do not share a currency. This is the usual configuration in emerging markets: the government's foreign-law bonds are denominated in dollars, while its domestic-law bonds pay local currency: 1^{LC} per unit of face under repayment and $\nu_D\,1^{LC}$ in default. Every other feature of case 1 carries over. The same p, ν_D, ν_F, and $r^{\$}$ price both bonds, default still

strikes the two instruments together, and the only new object is the exchange rate. Write $\mathcal{E}$ for the spot rate in local currency per dollar, so that a depreciation is a rise in $\mathcal{E}$. When the bonds mature the rate is $\mathcal{E}'_R$ if the government repays and $\mathcal{E}'_D$ if it defaults, each taken as known conditional on the state. Write $\mathcal{F}$, quoted the same way, for the forward rate contracted today, when the spot rate is $\mathcal{E}$.[2]

The two bonds cannot be compared directly. Converting the local-currency price at today's spot rate does not help: that gives a dollar figure, but the claim still *promises* local currency, so its yield and a dollar yield are not comparable. The comparison requires a claim that promises dollars. Assume a forward on the local currency is available, whether from covered interest parity or from a traded cross-currency swap (Du and Schreger, 2016). Nothing below depends on which.

The forward rate is pinned down by a single no-arbitrage restriction. A unit of local currency is worth $1^{LC}/\mathcal{E}'_R$ dollars under repayment and $1^{LC}/\mathcal{E}'_D$ dollars in default. A forward costs nothing to enter, so it must be worth zero at inception: the $\mathcal{F}$ units of local currency the investor commits to deliver have to be worth, in expectation, exactly the one dollar she receives for them. Under risk-neutral pricing that is

$$\frac{1^{LC}}{\mathcal{F}} = p\left(\frac{1^{LC}}{\mathcal{E}'_D}\right) + (1-p)\left(\frac{1^{LC}}{\mathcal{E}'_R}\right). \tag{23}$$

Write $1-\alpha \equiv \mathcal{F}/\mathcal{E}'_D$, so that α measures the depreciation upon default in excess of what the forward priced.[3] Sovereign defaults coincide with large depreciations, the Twin Ds pattern (Reinhart, 2002; Na et al., 2018). Measured against the no-default state the currency is weaker in default, $\mathcal{E}'_D > \mathcal{E}'_R$ (Mano, 2013), so $\alpha > 0$.

Now build the synthetic foreign-currency domestic-law bond. The investor buys $\mathcal{F}$ units of face of the local-currency bond and sells $\mathcal{F}$ units of local currency forward against one dollar. If the government repays, the bond delivers exactly the $\mathcal{F}$ units of local currency the forward demands, and she collects her dollar whatever the exchange rate turns out to be.

Default is where the construction slips. The forward is written on the face, but the government pays only ν_D of it, so the investor must buy the missing local currency at the

[2] $\mathcal{F}$ is a rate, local currency per dollar. As is usual we also treat it as a bare number.

[3] Given $\delta \equiv \mathcal{E}'_D/\mathcal{E}'_R > 1$, the fair-forward condition (23) makes α a function of p, since p prices the forward: $\alpha(p) = 1 - 1/\left[\delta + p\,(1-\delta)\right]$, so $\partial\alpha/\partial p = -\alpha\,(1-\alpha)/(1-p) < 0$. As default becomes more likely the forward is priced closer to $\mathcal{E}'_D$, which is why α falls.

post-default rate $\mathcal{E}'_D$ before she can deliver. Her dollar payoff is

$$\left[1-(1-\nu_D)\,\frac{\mathcal{F}}{\mathcal{E}'_D}\right]1^{\$}.$$

She still collects the dollar in full, because she delivers the whole face either way; that is the first term. The second is what the shortfall costs her. She owes the forward the whole face, the bond delivers ν_D of it, and she buys the remaining $(1-\nu_D)$ at the post-default rate. The whole face would cost $\mathcal{F}/\mathcal{E}'_D$ of a dollar at that rate, so the part she has to buy costs $(1-\nu_D)\mathcal{F}/\mathcal{E}'_D$.

Using $1-\alpha = \mathcal{F}/\mathcal{E}'_D$ and discounting the two payoffs, the bond prices at

$$\widetilde{q}^{\$}_D = \frac{1}{1+r^{\$}}\Big[(1-p)\,1^{\$} + p\,\big(\nu_D + \alpha\,(1-\nu_D)\big)1^{\$}\Big]. \tag{24}$$

The foreign-law bond follows (20), so the measured premium is

$$q^{\$}_F - \widetilde{q}^{\$}_D = \underbrace{\frac{p\,(\nu_F-\nu_D)\,1^{\$}}{1+r^{\$}}}_{\text{legal premium}} - \underbrace{\frac{p\,\alpha\,(1-\nu_D)\,1^{\$}}{1+r^{\$}}}_{\text{quanto}}, \tag{25}$$

The quanto term is the present value of the currency gain $\alpha\,(1-\nu_D)\,1^{\$}$ collected in default. It raises the dollar price of the synthetic bond, so the measured premium falls short of the legal premium in (21), and is itself negative once α exceeds $\alpha^\star \equiv (\nu_F-\nu_D)/(1-\nu_D)$.

Differentiating (25) gives

$$\frac{\partial(q^{\$}_F - \widetilde{q}^{\$}_D)}{\partial p} = \frac{(\nu_F-\nu_D)\,1^{\$}}{1+r^{\$}} - \frac{\bar{\alpha}\,(1-\nu_D)\,1^{\$}}{1+r^{\$}}, \qquad \bar{\alpha} \equiv \alpha + p\,\frac{\partial \alpha}{\partial p}. \tag{26}$$

Because the forward is priced to break even, the α the investor gains in default is offset in expectation by what she loses in repayment. A rise in p makes that gain more likely to be collected, but it also reprices the forward toward $\mathcal{E}'_D$ and so shrinks the gain itself, and $\bar{\alpha}$ combines the two.

Set (26) against (22). The true premium responds to credit risk by $(\nu_F-\nu_D)\,1^{\$}/(1+r^{\$})$, which is positive. The converted premium responds by the same quantity less $\bar{\alpha}\,(1-\nu_D)\,1^{\$}/(1+r^{\$})$, so its slope is negative whenever $\bar{\alpha}$ exceeds $\alpha^\star$.

A back-of-the-envelope calculation at the recoveries in Erce et al. (2024), $\nu_F = 0.67$ on foreign-law bonds against $\nu_D = 0.60$ on domestic-law bonds, gives $\alpha^\star = 0.07/0.40 =$

0.175. Currency-denominated CDS put the depreciation upon default at 36% (Mano, 2013; Du and Schreger, 2016), which fixes $\delta = 1.56$;[4] at $p \to 0$ the $p\,\partial\alpha/\partial p$ term drops out and $\bar{\alpha}$ equals that same 0.36. Substituting into (26) with $r^{\$} = 3\%$, the recovery term is $0.07/1.03 = 0.068$ and the quanto term is $0.36 \times 0.40/1.03 = 0.140$, so the slope is -0.072 per unit of p. A one-point rise in the default probability is $\Delta p = 0.01$, which moves the premium down by 7.2 basis points of face, against 6.8 up from the recovery advantage alone. The slope turns positive again only above a default probability of a third.

The quanto term loads on the same default probability that identifies the premium, so it survives bond fixed effects and first differencing, and a negative estimate is consistent with a substantial legal premium as easily as with none. Removing it requires a design that contracts no forward, which is that of case 3.

Case 3: both bonds in the same currency. The failure documented in case 2 originates in one feature of the construction. The forward rate is contracted today, while the quantity of local currency the investor receives is realized next period. Equation (23) states the consequence: a single rate $\mathcal{F}$ must satisfy one restriction spanning both states, so it cannot equal the state-contingent value of local currency in either, and the discrepancy in the default state is α. That discrepancy is strictly positive under the Twin Ds evidence, and through $\bar{\alpha}$ it enters the estimated slope with the opposite sign to the legal premium. Any design in which a forward is contracted inherits this problem.

We therefore use a comparison in which no forward is contracted at any stage. Several emerging-market governments have issued foreign-law bonds denominated in their own currency, so that a foreign-law and a domestic-law claim on the same government promise identical cash flows in identical units and differ only in the law that governs them. Section 2.4 describes these instruments. The investor is unchanged from case 1, pricing in dollars and discounting at $1 + r^{\$}$; what changes is that the bonds no longer pay in her unit of account.

Take a law-j bond of unit local-currency face. It pays 1 unit of local currency if the government repays, which occurs with probability $1 - p$, and ν_j units of local currency if it defaults, which occurs with probability p. Converting each payoff into dollars at the exchange rate prevailing when it is received, the bond is worth $1^{LC}/\mathcal{E}'_R$ dollars in repayment and $\nu_j\, 1^{LC}/\mathcal{E}'_D$ dollars in default. Weighting each state by its probability and discounting

[4] Du and Schreger (2016) measure the depreciation upon default as the fall in the currency's dollar value. Suppose the rate is 10 pesos per dollar under repayment, so a peso is worth \$0.10. A 36% loss of value leaves it worth \$0.064, at which a dollar buys $1/0.064 = 15.625$ pesos. The rate has gone from 10 to 15.625, so $\delta = 1.56$: the currency lost 36% of its value while the rate rose 56%.

gives

$$q_j^{\$} \;=\; \frac{1}{1+r^{\$}}\left[(1-p)\left(\frac{1^{LC}}{\mathcal{E}'_R}\right)+p\left(\frac{\nu_j\,1^{LC}}{\mathcal{E}'_D}\right)\right]. \tag{27}$$

Both bonds promise the same cash flow in repayment, so the repayment term is common to them and cancels in the difference:

$$q_F^{\$}-q_D^{\$} \;=\; \frac{p}{1+r^{\$}}\left(\frac{\nu_F\,1^{LC}}{\mathcal{E}'_D}-\frac{\nu_D\,1^{LC}}{\mathcal{E}'_D}\right) \;=\; \frac{p\,(\nu_F-\nu_D)\,1^{LC}}{(1+r^{\$})\,\mathcal{E}'_D}. \tag{28}$$

Equation (28) is stated per unit of local-currency face, so it carries the scale of the currency the bond is written in and its magnitude is not comparable across governments. We therefore work with prices expressed in local currency. Define

$$\omega \;\equiv\; \frac{\mathcal{E}}{\mathcal{E}'_D},$$

the fraction of its current dollar value that one unit of local currency retains in default. Evaluating (28) at the spot rate,

$$q_F^{LC}-q_D^{LC} \;=\; \mathcal{E}\left(q_F^{\$}-q_D^{\$}\right) \;=\; \frac{p\,(\nu_F-\nu_D)\,1^{LC}}{1+r^{\$}}\,\omega, \tag{29}$$

$$\frac{\partial\,(q_F^{LC}-q_D^{LC})}{\partial p} \;=\; \frac{(\nu_F-\nu_D)\,1^{LC}}{1+r^{\$}}\,\omega. \tag{30}$$

The exchange rate enters (30) through ω alone, and $\omega = 1/(g\,\delta)$, where $g \equiv \mathcal{E}'_R/\mathcal{E}$ is the depreciation expected absent default and δ the jump upon default of case 2. It is strictly positive because exchange rates are positive in every state, so the slope has the sign of $\nu_F - \nu_D$ whatever the value of ω. Equation (30) is a difference in prices. The yield gap that we estimate in Section 2.4 carries only the jump at default, $1/\delta$, because the expected depreciation g enters both yields and cancels; Appendix A.4 derives this and measures δ.

The design also has a sharp null. On an instrument sharing the treatment's currency, settlement and clientele but no difference in governing law, $\nu_F = \nu_D$ and (30) is exactly zero for any ω. Conversion returns a strictly negative number instead, so under it a true zero and a masked premium look alike. Appendix A.11 reports this test.

Set (30) against (22) and (26). At the recoveries and depreciation used above, a one-point rise in the default probability moves the true premium of case 1 up 6.8 basis points of face and the converted premium of case 2 down 7.2. At a 10% depreciation absent default, $\omega =$

0.58 and the same-currency premium moves up 4.0.[5] The exchange rate multiplies the same-currency premium and subtracts from the converted one. A positive factor cannot change a sign, so the same-currency comparison gives the sign of the recovery advantage for any ω, while conversion can report it with the wrong sign. We use the same-currency comparison throughout.

From the model to the data. The model compares two claims with identical cash flows. Actual bonds carry coupons and maturities, no domestic-law bond matches a given foreign-law one, and the currencies at issue are not deliverable offshore. None of this need break the comparison, but each has to be dealt with.

From prices to yields, and maturity. Case 1 measures the premium as a yield gap. Prices there are in dollars throughout. In the one-period case, with $y_j^{\$} = 1/q_j^{\$} - 1$,

$$\Pi^{y} \equiv y_D^{\$} - y_F^{\$} = \frac{1}{q_D^{\$}} - \frac{1}{q_F^{\$}} = \frac{q_F^{\$} - q_D^{\$}}{q_D^{\$}\, q_F^{\$}} = \frac{\Pi^{\$}}{q_D^{\$}\, q_F^{\$}}, \tag{31}$$

so the yield premium is the price premium scaled by the positive factor $1/(q_D^{\$} q_F^{\$})$, and carries its sign. Since $\partial q_j^{\$}/\partial p = (\nu_j - 1)/(1 + r^{\$}) < 0$, both prices fall as p rises, so the factor rises with p and the yield premium widens wherever the price premium does.

The multi-period case. A law-j zero-coupon bond of maturity n and unit face pays 1 at n under repayment and the recovery ν_j at the date of first default otherwise, so under risk-neutral pricing[6]

$$q_j^{(n)} = \sum_{k=1}^{n} (1-p)^{k-1}\, p\, \nu_j\, \frac{1}{(1+r^{\$})^k} \;+\; \frac{(1-p)^n}{(1+r^{\$})^n}. \tag{32}$$

Differencing across the two laws, the common repayment payoff cancels:

$$\Pi^{\$(n)} = p\,(\nu_F - \nu_D)\, A_n(p), \qquad A_n(p) \equiv \sum_{k=1}^{n} \frac{(1-p)^{k-1}}{(1+r^{\$})^k} > 0, \tag{33}$$

The factor $p\, A_n(p) = \sum_{k=1}^{n} p\,(1-p)^{k-1}(1+r^{\$})^{-k}$ is the present value of one unit paid at the first default within n periods, and it rises with p, so $\partial \Pi^{\$(n)}/\partial p > 0$ exactly as in the

[5] $\delta = 1.56$ from the 36% of case 2. The slope is 4.3 basis points with no depreciation absent default and 3.5 at 25%.

[6] The per-period default hazard p is again common to the two instruments, as in case 1.

one-period case (22). Going from n to $n+1$ adds the positive term $(1-p)^n(1+r^\$)^{-(n+1)}$ to A_n, so A_n rises with n and the premium rises with maturity in price space.

Maturity in yield space. Yields are annualized, $y_j^{(n)} = \left(1/q_j^{(n)}\right)^{1/n} - 1$, so the yield premium is

$$\Pi^{y(n)} = \left(q_D^{(n)}\right)^{-1/n} - \left(q_F^{(n)}\right)^{-1/n} \simeq \frac{\Pi^{\$(n)}}{n\left(q^{(n)}\right)^{(n+1)/n}},$$

with $q^{(n)}$ between the two prices. The numerator rises with n, and the divisor carries the two opposing forces: the horizon n over which the yield is annualized, which shrinks the gap, and the price $q^{(n)}$, which falls with n and magnifies it. At $\nu_F = 0.67$, $\nu_D = 0.60$, $r^\$ = 3\%$ and $p = 0.05$ the yield premium is 37, 38, 38, 35 and 30 basis points at $n = 1, 5, 10, 20, 30$, rising and then falling. The slope, by contrast, is nearly invariant to p: at the same calibration it falls from 0.077 to 0.073 as the annual hazard rises from one to eight percent. This is why residual maturity enters the empirical specification (4) only as a control, with no sign restriction.

A.4 Depreciation at default and the yield-gap estimates

We measure the depreciation of the local currency at default from two CDS contracts on the same government, and we show that this depreciation is the only exchange-rate term that scales the yield gap. A one-period model delivers three results: the ratio of the dollar CDS spread to the local-currency CDS spread equals the depreciation at default; the yield gap between the two laws is divided by that depreciation; and the gap can be corrected on either side of the regression, but on one side only.

Investors are risk neutral in dollars and discount at the dollar rate r. The government defaults next period with probability p. We quote the exchange rate in local currency per dollar: $\mathcal{E}$ today, $\mathcal{E}'_R$ next period if the government repays and $\mathcal{E}'_D$ if it defaults. The depreciation at default is $\delta \equiv \mathcal{E}'_D/\mathcal{E}'_R$, the default-state rate relative to the repayment-state rate, and the ordinary depreciation is $g \equiv \mathcal{E}'_R/\mathcal{E}$.

A dollar CDS with one dollar of notional pays $1-\nu_C$ at default, where ν_C is the recovery set at the ISDA auction, and the buyer pays the spread $s^\$$ if the government repays. Its value is zero at inception, so $(1-p)\,s^\$ = p\,(1-\nu_C)$. A local-currency CDS on the same government has $\mathcal{E}$ units of local currency as notional and settles at the same auction and price, so it pays the same fraction $1-\nu_C$ of its notional, in local currency, against the spread s^{LC}. Valued in dollars, $(1-p)\,s^{LC}\,\mathcal{E}/\mathcal{E}'_R = p\,(1-\nu_C)\,\mathcal{E}/\mathcal{E}'_D$. Dividing the first

condition by the second,

$$\delta = \frac{s^{\$}}{s^{LC}}. \tag{34}$$

The default probability, the recovery, today's exchange rate and the ordinary depreciation all cancel, and the exchange-rate jump at default remains: the local-currency protection pays in a currency that has lost value, and it is cheaper by that jump. Because δ is a jump at the default date, one value applies to bonds of every maturity.

A law-j bond pays one unit of local currency if the government repays and ν_j units if it defaults. Its price in local currency is

$$q_j = \frac{\mathcal{E}}{1+r}\left[\frac{1-p}{\mathcal{E}'_R} + \frac{p\,\nu_j}{\mathcal{E}'_D}\right] = \frac{(1-p) + p\,\nu_j/\delta}{g\,(1+r)}.$$

With one-period yields $y_j = 1/q_j - 1$, and for a small default probability, the yield gap is

$$y_D - y_F = \frac{q_F - q_D}{q_D\,q_F} \simeq g\,(1+r)\,\frac{p\,(\nu_F - \nu_D)}{\delta}. \tag{35}$$

The factor $g(1+r)$ is approximately the gross local-currency interest rate, close to one. The ordinary depreciation enters the gap only through the interest rate, and the depreciation at default divides the recovery advantage. Over n periods the same holds with the annual default probability in place of p, since the jump occurs once.

With $s^{\$} \simeq p\,(1-\nu_C)$, the slope of the gap on the dollar CDS is $(\nu_F - \nu_D)/[\delta\,(1-\nu_C)]$. Two regressions remove δ: the gap multiplied by δ_t on the dollar CDS, and the raw gap on the local-currency CDS, since $s^{LC} = s^{\$}/\delta$. Both slopes equal $(\nu_F - \nu_D)/(1-\nu_C)$. Multiplying the gap by δ_t and regressing it on the local-currency CDS removes δ twice.

Du and Schreger (2016) model depreciation at default in the same way: the local currency depreciates by a fraction α in the default state relative to the nondefault state, so that $\alpha = 1 - 1/\delta$. The depreciation priced at default matches the evidence that defaults coincide with large depreciations (Reinhart, 2002; Na et al., 2018).

We compute δ_t daily from Markit's 5-year par spreads, after two screens. The first drops days on which the local-currency spread equals the dollar spread: until a local-currency curve is quoted, Markit republishes the dollar curve under the local-currency ticker. The second drops days with $\delta_t < 1$. Six of the twelve governments in our sample have a quoted local-currency CDS. Table 14 reports δ for five of them over 2015–2025; across them, the local currency loses between 35 and 50 percent of its dollar value at default.

Brazil's local-currency CDS is quoted only until May 2015.[7] Figure 6 plots the daily series.

Table 14. Depreciation at Default

	δ	s.e.	Median	Loss of dollar value (%)	Days
Colombia	1.55	0.006	1.61	35	1,999
Mexico	1.55	0.007	1.55	36	2,231
Russia	1.61	0.004	1.63	38	1,862
South Africa	2.01	0.009	1.94	50	2,610
Turkey	1.54	0.005	1.53	35	2,627

Notes: δ_t is the 5-year dollar CDS spread over the 5-year local-currency CDS spread on the same government, Markit par composites, January 2015 to June 2025 (Mexico to December 2024, Russia to February 2022). δ is the mean over days that pass both screens, with a Newey–West standard error. The loss of dollar value is $1 - 1/\delta$.

Table 15 reports the three regressions for Colombia and Russia, the two governments in our yield-gap sample with a quoted local-currency CDS over their estimation windows. Each column re-estimates the country's baseline regression, changing only the dependent variable or the CDS. In Colombia, multiplying the gap by δ_t raises the slope from 0.23 to 0.36, and the regression on the local-currency CDS gives 0.33. In Russia, the slope rises from 0.08 to 0.11; the regression on the local-currency CDS gives 0.03, with daily changes in the two CDS spreads correlated at 0.56, against 0.93 in Colombia.

Table 15. The Yield Gap and Depreciation at Default

	Gap on dollar CDS		Gap $\times$ δ_t on dollar CDS		Gap on local-currency CDS	
	Slope	s.e.	Slope	s.e.	Slope	s.e.
Colombia	0.227	0.038	0.356	0.060	0.331	0.057
Russia	0.075	0.030	0.109	0.046	0.027	0.017

Notes: Each country's first-difference regression of Table 4, without the VIX control, with HC1 standard errors, on days with a screened δ_t. Colombia: 2017–2020, mean δ_t 1.57, 1,373 observations. Russia: 2014–2017, mean δ_t 1.56, 652 observations. Brazil's local-currency CDS covers 15 percent of its window, and Paraguay has none.

[7] Over Brazil's quoted span, 2010–2015, δ averages 1.37. Markit has no local-currency curve for Hungary, Peru and Poland, where the local-currency ticker carries the dollar curve, and none for Chile; we have not obtained local-currency quotes for Paraguay and Uruguay.

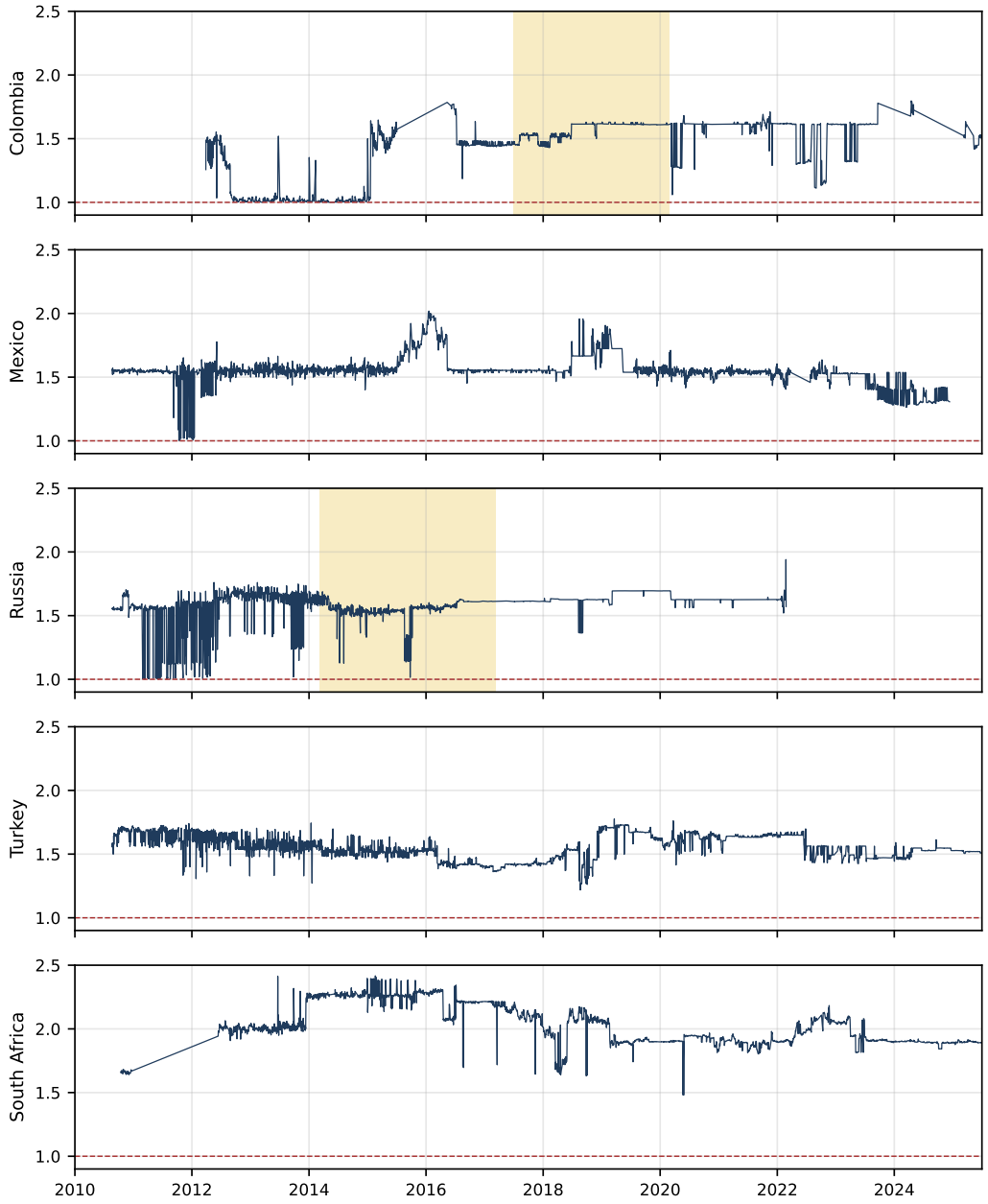


Figure 6. Daily Depreciation at Default

Notes: δ_t, the 5-year dollar CDS spread over the 5-year local-currency CDS spread, Markit par composites, on days that pass both screens. Shaded: the estimation windows of the yield gap for Colombia and Russia. Dashed line: $\delta = 1$.

A.5 International investors in domestic-law debt

The model prices both bonds with the same risk-neutral international investors. This requires that foreign investors can buy and hold the domestic-law debt, and that they do. Section 2.4 restricts each sample to the years in which they can buy, hold and sell it without restrictions or punitive taxes. Table 16 reports the share of each government's domestic-law marketable debt held by nonresidents, from national sources: the central bank, the treasury or the debt office. The first two columns are means over the full period of each source, and the last two are means over the estimation window of the yield gap (Table 3). The columns marked "all holders" divide nonresident holdings by the debt held by every holder, including the central bank and public institutions. The columns marked "net of public" remove the holdings of the central bank and of public institutions and funds from the denominator, which leaves the debt that trades among private investors.

Table 16. Nonresident Holdings of Domestic-Law Debt

		Full period		Estimation window	
	Source, period	All holders	Net of public	All holders	Net of public
Brazil	Tesouro Nacional, 2007–2026	11.4	12.0	15.2	16.1
Chile	Banco Central, 2012–2026	9.5	—		
Colombia	MinHacienda, 2010–2026	17.3	20.0	25.6	29.2
Hungary	ÁKK, 2004–2026	29.1	30.7		
Mexico	Banxico, 2004–2026	20.9	21.8		
Paraguay	MEF, 2025–2026	3.5	—	—	
Peru	MEF, 2007–2026	41.4	42.6	43.6	45.2
Poland	Ministry of Finance, 2004–2026	25.4	27.5		
Russia	Bank of Russia, 2011–2026	19.1	24.3	22.9	32.7
South Africa	National Treasury, 2006–2026	31.6	43.7		
Turkey	Treasury, 2004–2026	12.2	16.6		
Uruguay	Debt office, 2024–2026	2.2	—		

Notes: Percent of domestic-law marketable government debt held by nonresidents, monthly or quarterly means. Net of public removes: Brazil, government funds (the central bank is already excluded); Colombia, the central bank and the core public sector; Hungary, Poland and Turkey, the central bank and the government sector or public banks; Mexico, Banxico's holdings; Peru, public funds; Russia, the Bank of Russia and direct-placement tranches; South Africa, the government pension fund. Paraguay: two observations; Uruguay: implied from the debt office's composition reports. Estimation windows: Brazil 2014–2020, Colombia 2017–2020, Russia 2014–2017, Peru 2017–2025 excluding March 2020 to July 2021.

A.6 Defaults on both classes of bonds

The model assumes that the government defaults on both classes of bonds at once. We assess this assumption with the database of Erce et al. (2024), January 2026 release, which records sovereign default and restructuring events on domestic and external debt. Our

model concerns marketable bonds held by investors who can buy bonds under both laws, so we select the episodes that match this setting in three steps. Table 17 reports the events in the database, Table 18 the episodes after the first two steps, and Table 19 the episodes after the third.

First, we keep bonds that private investors bought in the market. The database records 297 events in 84 countries, of which 184 are bank loans and commercial credit and 113 are bonds. Using the authors' companion volume of sovereign histories, which names the instrument in each episode, we keep the 90 bond events in which an instrument traded in the secondary market. Grouping these events into episodes as the authors do gives 43 episodes in 31 countries.

Second, we keep episodes in which bonds under both laws were outstanding. We measure the existence of foreign-law bonds with Dealogic, as bonds issued before the episode and maturing after it, and the existence of a domestic bond market from the authors' episode narratives. Bonds under both laws were outstanding in 27 of the 43 episodes. In 12 of them, both classes of bonds were restructured; in 8, only domestic-law bonds; and in 7, only foreign-law bonds.

Third, among the 15 episodes in which only one class was restructured, we keep those in which foreign investors could hold the domestic-law debt. In 11 of them, exchange controls or terms that applied only to residents limited the domestic-law debt to resident holders. Among the remaining episodes, both classes of bonds were restructured in 12 of 14 under a strict classification and in 12 of 16 under a lenient one.

Table 17. Default Events in the Database

	Events	Percent	US$ billion
All events	297	100.0	1,671.6
Bank loans and commercial credit	184	62.0	736.9
Bonds	113	38.0	934.8
	Domestic	Foreign	Total
Bond events: bought by private investors	59	31	90
Bond events: other instruments	23	0	23

Notes: Events in Erce et al. (2024), January 2026 release. A bond event counts as bought by private investors if any instrument in it was priced in the secondary market. Domestic and foreign refer to the market of issuance.

Table 18. Default Episodes on Bonds Bought by Private Investors

	All episodes		Bonds under both laws outstanding	
	Episodes	Percent	Episodes	Percent
Both classes hit	12	27.9	12	44.4
Domestic-law bonds only	18	41.9	8	29.6
Foreign-law bonds only	13	30.2	7	25.9
Total	43	100.0	27	100.0

Notes: Episodes as published by the authors. The episodes that hit both classes are Argentina 2001, Barbados 2018, Côte d'Ivoire 2011, Dominica 2003, Ecuador 1997, Greece 2011, Grenada 2004 and 2013, Russia 1998, St Kitts and Nevis 2011, Ukraine 1998 and Uruguay 2002.

Table 19. Episodes in Which Foreign Investors Could Hold Domestic-Law Debt

	Strict		Lenient	
	Episodes	Percent	Episodes	Percent
Both classes hit	12	85.7	12	75.0
One class restructured	2	14.3	4	25.0
Total	14	100.0	16	100.0

Notes: Of the 15 episodes in which only one class was restructured, 11 are dropped because the domestic-law debt was limited to resident holders: Belize 2006, 2012 and 2016, Brazil 1986, Cyprus 2013, Jamaica 2010 and 2013, Mozambique 2015 and 2016, Seychelles 2008 and Venezuela 1998. Strict keeps Turkey 1999 and Panama 1998; lenient also keeps Venezuela 2002 and Pakistan 1998.

A.7 Instruments and curves

Paraguay has no published zero curve for its domestic-law guaraní bonds, so we fit one each day. We use the eleven domestic-law Bonos del Tesoro that Bloomberg prices with BVAL, its evaluated price; between 9 and 11 of them are quoted on a given day. The bonds pay semiannual coupons, and we compute accrued interest on a 30/360 basis. Bloomberg's BVAL history for these bonds begins on 6 February 2025, which sets the start of Paraguay's sample.

On each day we fit the Nelson–Siegel curve

$$y(t) = \beta_0 + \beta_1 \frac{1 - e^{-t/\lambda}}{t/\lambda} + \beta_2 \left(\frac{1 - e^{-t/\lambda}}{t/\lambda} - e^{-t/\lambda} \right),$$

where $y(t)$ is the annually compounded zero yield at a horizon of t years. Given the curve, we price each bond by discounting each remaining cash flow at the zero yield of its own date, $P^{\text{model}} = \sum_k CF_k \left(1 + y(t_k)\right)^{-t_k}$. We choose $(\beta_0, \beta_1, \beta_2)$ to minimize

$$\sum_{\text{bonds}} \left(\frac{P^{\text{model}} - P^{\text{market}}}{\text{DV01}} \right)^2,$$

where P^{market} is the bond's dirty mid price and DV01 is its price change for a one-basis-point change in its yield. Dividing by DV01 turns each price error into a yield error, so long bonds do not dominate the fit. We drop bonds within 30 days of maturity and days with fewer than four quotes.

The decay parameter λ sets where the curvature term peaks. Each day we select λ from a grid between 0.15 and 3 years, and we use the 21-day rolling median of the daily choices. Of the alternatives we compared, this one keeps the fit within 25 percent of the best daily fit and moves the curve least from one day to the next. λ equals 3 on 278 of the 331 days.

We use the curve only within the maturities quoted that day, from the shortest remaining domestic bond to the longest. A foreign-law bond enters the sample on a given day only if its final payment falls within that range. Over the 331 days, the median root-mean-squared yield error of the fit is 7 basis points. We also refit the curve without one bond at a time and price the omitted bond on it. The median absolute error is 7 basis points when the omitted bond lies inside the range of the others and 16 basis points when it lies at either end; this is why we do not extrapolate. Fitting the curve separately to bid and to ask prices moves it by about 8 basis points at ten years and 12 at five years, measured as

half the distance between the two curves.

Figure 7 shows the fit on four dates. Each red point is a domestic-law bond's yield to maturity, plotted at its duration. For a coupon bond this yield is close to the zero yield at its duration, so the vertical distance from a point to the line approximates that bond's fitting error. The crosses place each foreign-law Global at its own duration on the fitted curve. The range of quoted maturities and the number of bonds change across dates as domestic bonds mature and new ones are issued, and the 2038 Global enters in February 2026.

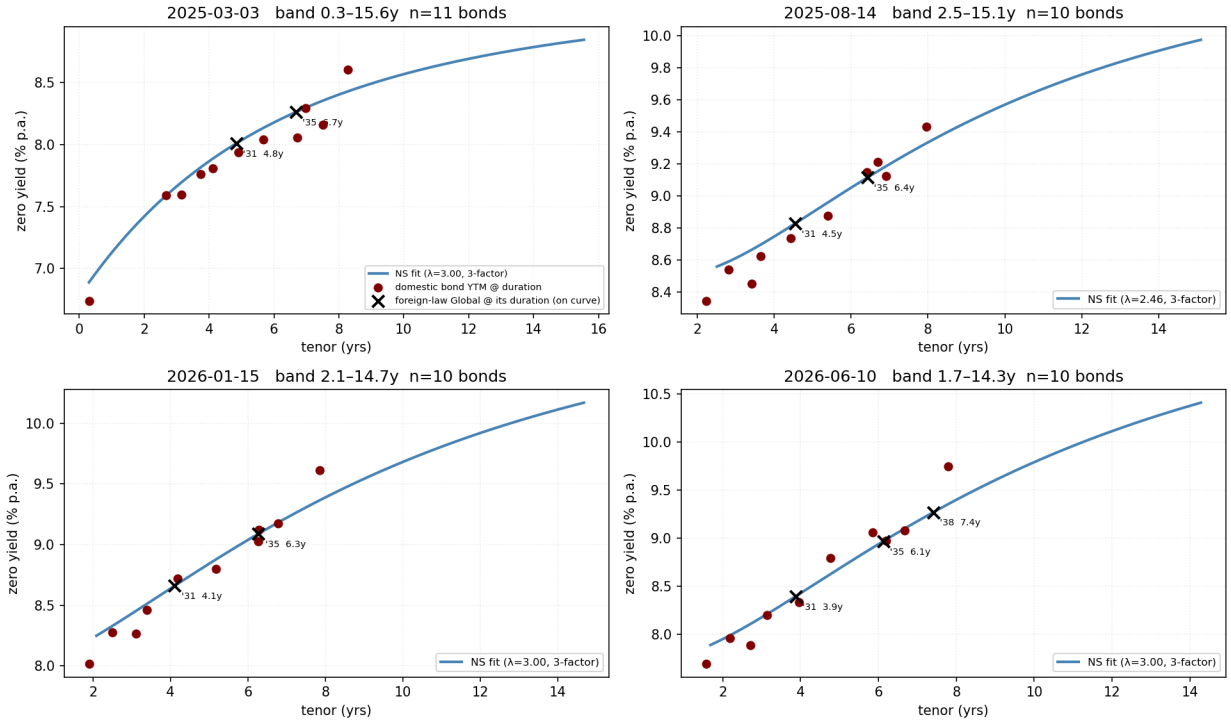


Figure 7. Paraguay's Fitted Domestic-Law Curve

Notes: Daily Nelson–Siegel zero curve fitted to the domestic-law Bonos del Tesoro on four dates. Points: each domestic-law bond's yield to maturity at its duration. Crosses: each foreign-law Global at its duration, on the fitted curve. Panel titles give the range of quoted maturities and the number of domestic-law bonds in the fit.

A.8 Stationarity, cointegration and inference

A.9 Robustness of the yield-gap estimates

Table 20 re-estimates equation (4) under alternative specifications, country by country, without and with the change in the VIX. The first rows report the Driscoll–Kraay t-statistic of the baseline, drop the liquidity control, drop the maturity control, and keep only differences between consecutive trading days. The window rows extend Brazil's sample past

the COVID break, from July 2020 or from July 2021 to December 2025; extend Colombia's to 2017–2025 and to 2015–2025, excluding the COVID period; exclude the Russian crisis months from November 2014 to March 2015; and extend Peru's sample to 2017–2025 including the COVID period. The last rows drop one bond at a time.

Table 20. The Yield Gap and the CDS Spread: Robustness

	Without ΔVIX		With ΔVIX	
	Slope	t	Slope	t
Brazil				
Baseline	0.669^{***}	(10.65)	0.694^{***}	(9.72)
Driscoll–Kraay t-statistic	0.669^{***}	(7.37)	0.694^{***}	(6.80)
Without the liquidity control	0.668^{***}	(10.62)	0.694^{***}	(9.68)
Without the maturity control	0.669^{***}	(10.66)	0.694^{***}	(9.74)
One-trading-day differences	0.703^{***}	(8.96)	0.738^{***}	(8.21)
Adding July 2020 to December 2025	0.590^{***}	(10.83)	0.631^{***}	(10.21)
Adding July 2021 to December 2025	0.628^{***}	(11.28)	0.676^{***}	(10.64)
Without the 2022 bond	0.657^{***}	(7.97)	0.686^{***}	(7.27)
Without the 2028 bond	0.681^{***}	(10.29)	0.706^{***}	(9.42)
Without the 2024 bond	0.659^{***}	(7.56)	0.682^{***}	(6.87)
Colombia				
Baseline	0.221^{***}	(5.74)	0.225^{***}	(4.83)
Driscoll–Kraay t-statistic	0.221^{***}	(4.35)	0.225^{***}	(3.81)
Without the liquidity control	0.221^{***}	(5.74)	0.225^{***}	(4.83)
Without the maturity control	0.221^{***}	(5.74)	0.225^{***}	(4.83)
One-trading-day differences	0.104^{***}	(3.16)	0.117^{***}	(2.93)
2017–2025, excluding the COVID period	0.231^{***}	(3.25)	0.250^{***}	(2.90)
2015–2025, excluding the COVID period	0.187^{***}	(3.47)	0.203^{***}	(3.10)
Without the 2027 bond	0.265^{***}	(5.25)	0.260^{***}	(4.32)
Without the 2021 bond	0.205^{***}	(4.55)	0.229^{***}	(4.00)
Without the 2023 bond	0.193^{***}	(4.31)	0.188^{***}	(3.55)
Paraguay				
Baseline	0.157^{**}	(2.03)	0.169^{**}	(2.10)
Driscoll–Kraay t-statistic	0.157^{**}	(2.20)	0.169^{**}	(2.16)
Without the liquidity control	0.222^{***}	(2.83)	0.231^{***}	(2.81)
Without the maturity control	0.158^{**}	(2.03)	0.170^{**}	(2.10)
One-trading-day differences	0.109	(0.96)	0.108	(0.90)
Without the 2031 bond	0.298^{***}	(2.94)	0.364^{***}	(3.43)
Without the 2035 bond	0.088	(0.87)	0.059	(0.58)
Without the 2038 bond	0.117	(1.49)	0.123	(1.54)
Russia				
Baseline	0.079^{***}	(2.64)	0.073^{**}	(2.30)
Driscoll–Kraay t-statistic	0.079^{***}	(3.38)	0.073^{***}	(2.83)
Without the liquidity control	0.098^{***}	(2.80)	0.097^{**}	(2.54)
Without the maturity control	0.079^{***}	(2.64)	0.072^{**}	(2.30)
One-trading-day differences	0.028	(0.76)	0.018	(0.46)
Excluding November 2014 to March 2015	0.059^{*}	(1.75)	0.049	(1.38)
Peru (placebo)				
Baseline	-0.028	(-1.46)	-0.017	(-0.76)
Driscoll–Kraay t-statistic	-0.028	(-1.08)	-0.017	(-0.58)
Without the liquidity control	-0.027	(-1.42)	-0.018	(-0.78)
Without the maturity control	-0.028	(-1.46)	-0.017	(-0.75)
One-trading-day differences	-0.038^{*}	(-1.93)	-0.046^{**}	(-2.05)
2017–2025, including the COVID period	-0.021	(-1.01)	-0.022	(-0.95)
Without the 2037 bond	-0.014	(-0.54)	0.011	(0.33)
Without the 2031 bond	-0.054^{***}	(-2.87)	-0.052^{**}	(-2.41)
Without the 2028 bond	-0.017	(-0.72)	-0.012	(-0.41)

Notes: Estimates of γ in equation (4), without and with the change in the VIX among the controls, on the days for which the VIX is available. t-statistics in parentheses use Huber–White standard errors, except in the Driscoll–Kraay row (Newey–West for Russia, which has one bond). Stars follow the t-statistic shown: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

A.10 Binned scatters by country and bond-by-bond estimates

A.11 Placebo: Peru's Global Depositary Notes

Peru's sol Global Depositary Notes give a placebo for the yield gap. Each note is a claim, issued by a depositary bank, on domestic-law Bonos Soberanos held in custody in Peru. New York law governs the deposit agreement, and the payment claim remains under Peruvian law, so both sides of the gap are domestic-law claims. Nonresidents can buy, hold and sell the Bonos Soberanos without restrictions, and held 40.9 percent of their outstanding stock at the end of 2024.[8]

We use the notes on the Bonos Soberanos maturing in 2028, 2031 and 2037, from April 2017 to December 2025, excluding March 2020 to July 2021, and apply the construction and the regression of Section 2.4. The slope of the gap on Peru's CDS spread is -0.017, with a Driscoll–Kraay t-statistic of -0.58 and 3,391 observations. Table 20 reports the alternative specifications.

A.12 The cost of issuing

We estimate how the fee varies with maturity and size with one regression,

$$\text{fee}_i = \beta_M \,\text{maturity}_i + \beta_S \,\text{size}_i + \alpha_{c(i)} + \tau_{t(i)} + \varepsilon_i,$$

where fee_i is the underwriting fee of bond i in basis points of face value, maturity_i its maturity at issuance in years, size_i its face value in US$ billion, and $\alpha_{c(i)}$ and $\tau_{t(i)}$ country and year effects. The sample is the New York-law bonds with a fee, without two retail notes that Mexico sold in 2006, and without the bonds below the 1st and above the 99th percentile of maturity: three bonds shorter than 5 years and four longer than 50 years. This leaves 360 bonds from ten countries. Standard errors are clustered by country, and because there are only ten clusters we also report p-values from a wild bootstrap. Neither coefficient is significant: $p = 0.53$ for maturity (0.61 with the bootstrap) and $p = 0.95$ for size (0.95), and the joint test that both are zero gives $p = 0.82$. With the seven bonds outside the percentiles included, the coefficient on maturity is 0.042, with $p = 0.30$. For Colombia, the same regression on its 49 New York-law bonds, with year effects and errors clustered by deal, gives $p = 0.90$ for maturity and $p = 0.39$ for size (Figure 8).

Only five countries have bonds with a recorded fee under laws other than New York law,

[8] Ministerio de Economía y Finanzas, *Informe Anual de Deuda Pública 2024* (May 2025).

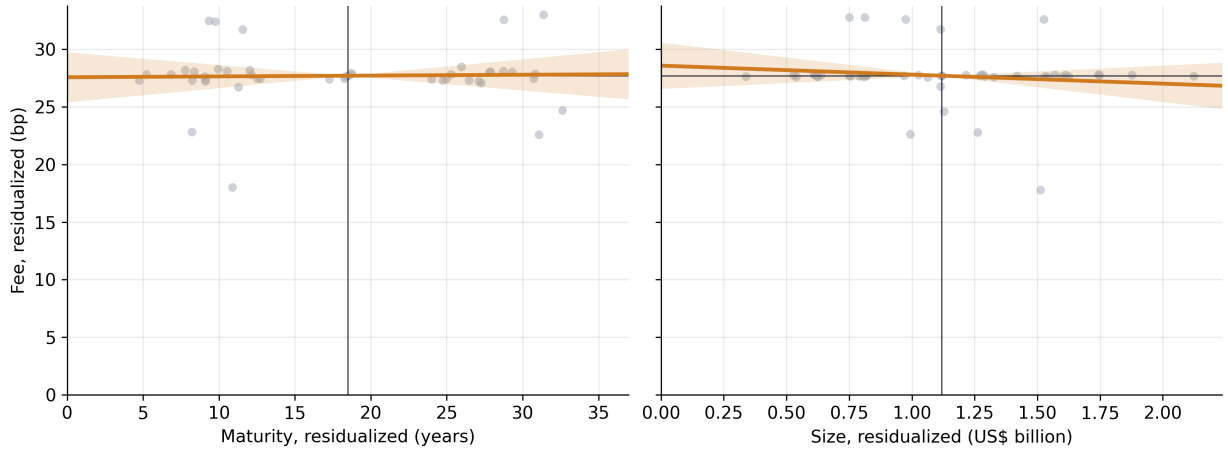


Figure 8. Underwriting Fee, Maturity and Size: Colombia

Notes: As Figure 2, for Colombia's 49 New York-law bonds, 2006–2024, with year effects; averages added back (fee 27.7 basis points, maturity 18.5 years, size US$1.1 billion). Shaded: 95 percent interval of the slope, with errors clustered by deal.

PROSPECTUS SUPPLEMENT
(TO PROSPECTUS DATED APRIL 15, 2021)

U.S. $2,000,000,000

Republic of Colombia

3.250% Global Bonds due 2032

The bonds will mature on April 22, 2032. The Republic of Colombia ("Colombia" or the "Republic") will pay interest on the bonds each April 22 and October 22, commencing on October 22, 2021. The bonds will be issued in denominations of U.S. $200,000 and integral multiples of U.S. $1,000 in excess thereof.

The bonds will be direct, general, unconditional, unsecured and unsubordinated external indebtedness of Colombia and will be backed by the full faith and credit of Colombia. The bonds will rank without any preference among themselves and equally with all other unsecured and unsubordinated external indebtedness of Colombia. It is understood that this provision shall not be construed so as to require Colombia to make payments under the bonds ratably with payments being made under any other external indebtedness.

Colombia may, at its option, redeem the bonds, in whole or in part, before maturity, on not less than 10 nor more than 60 days' notice on the terms described under "Description of the Bonds—Optional Redemption" in this prospectus supplement. The bonds will not be entitled to the benefit of any sinking fund.

The bonds will be issued under an indenture and constitute a separate series of debt securities under the indenture. The indenture contains provisions regarding future modifications to the terms of the bonds that differ from those applicable to Colombia's outstanding public external indebtedness issued prior to January 28, 2015. Under these provisions, which are described beginning on page 7 of the accompanying prospectus, Colombia may amend the payment provisions of any series of debt securities (including the bonds) and other reserve matters listed in the indenture with the consent of the holders of: (1) with respect to a single series of debt securities, more than 75% of the aggregate principal amount of the outstanding debt securities of such series; (2) with respect to two or more series of debt securities, if certain "uniformly applicable" requirements are met, more than 75% of the aggregate principal amount of the outstanding debt securities of all series affected by the proposed modification, taken in the aggregate; or (3) with respect to two or more series of debt securities, more than 66 2/3% of the aggregate principal amount of the outstanding bonds of all series affected by the proposed modification, taken in the aggregate, and more than 50% of the aggregate principal amount of the outstanding debt securities of each series affected by the proposed modification, taken individually.

Application will be made to list the bonds on the official list of the Luxembourg Stock Exchange and to trade them on the Euro MTF Market of the Luxembourg Stock Exchange.

See "Risk Factors" beginning on page S-7 to read about certain risks you should consider before investing in the bonds.

Section 309B(1)(c) of the Securities and Futures Act (Chapter 289 of Singapore) Notification

The bonds are prescribed capital markets products (as defined in the Securities and Futures (Capital Markets Products) Regulations 2018).

Neither the Securities and Exchange Commission, referred to as the SEC, nor any other regulatory body has approved or disapproved of these securities or determined if this prospectus supplement or the accompanying prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

	Per bond	Total
Public offering price(1)	99.032%	U.S. $1,980,640,000
Underwriting discount	0.200%	U.S. $ 4,000,000
Proceeds, before expenses, to Colombia	98.832%	U.S. $1,976,640,000

(1) Purchasers will also be required to pay accrued interest, if any, from April 22, 2021, if settlement occurs after that date.

Source: Republic of Colombia, prospectus supplement (424B5) for the U.S.$2,000,000,000 3.250% Global Bonds due 2032, April 2021. SEC EDGAR accession 0001193125-21-124194, cover page. Highlight added.

Figure 9. The Cover of a Prospectus

Table 21. Issuer Expenses per Deal

Country	Prospectuses with an expense figure (1)	Expenses, median, US$ thousand (2)	Expenses, s.d., US$ thousand (3)	Share of the cost of a deal, % (4)
Brazil	29	185	121	5
Chile	30	150	106	20
Colombia	34	300	44	7
Hungary	7	175	71	5
Mexico	40	250	88	4
Peru	14	265	73	17
Poland	11	200	79	6
South Africa	14	305	187	12
Turkey	1	—	—	—
Uruguay	20	120	40	10

Notes: Expenses the issuer pays outside the underwriting fee (legal, listing, agents), as estimated in each SEC prospectus, New York-law bonds, 2006–2024. Where the stated figure includes the fee, the fee is subtracted. Column (4) is the median, deal by deal, of expenses over the fee plus expenses.

Table 22. Underwriting Fee on Bonds under Other Foreign Laws

	Bonds (1)	Median (2)	Mean (3)	Mean, face-weighted (4)
Hungary	6	30.0	33.3	22.0
Mexico	18	26.0	26.6	25.8
Poland	21	35.0	38.8	35.3
Turkey	1	13.0	13.0	13.0
Uruguay	4	8.0	8.0	8.0

Notes: As in Table 5, for bonds issued under Japanese, English and Swiss law, 2006–2024.

Table 23. Underwriting Fee by Governing Law and Market

	All countries			Same country: median		
Law and market	Bonds	Median	Mean	Mexico	Poland	Hungary
New York law	368	15.0	16.5	19.0	14.0	18.0
English law, euro	8	14.0	14.2	—	14.5	10.0
Japanese law, yen	37	29.0	29.2	26.0	35.0	30.0
Swiss franc	4	75.0	81.2	—	75.0	75.0
English law, hand-collected	37	14.0	13.5	—	14.0	12.5

Notes: Underwriting fee in basis points of face value, Dealogic, 2006–2024. The last three columns give the median fee of each country's bonds in each market. The last row uses English-law fees read from Luxembourg and London exchange filings.

50 bonds in all, and three quarters of them are yen bonds issued under Japanese law. Table 23 compares the fee by law and market. English-law bonds in euros pay about the same fee as New York-law bonds, 14 against 15 basis points, and in Poland 14.5 against 14.0. Yen bonds pay about twice as much, and Swiss-franc bonds more. Among these 50 bonds, a 30-year bond pays 5.5 basis points less than a 10-year bond of the same country and year, with a 95 percent interval from -8.8 to -2.1.

A.13 How often and how much governments issue

Governments borrow abroad less often than at home, and in larger operations. Table 24 reports, for each country, how often the government issues under each law and how much it raises each time, by quarter, over 2006–2024. The share of quarters with no foreign-law issuance ranges from 0.26 in Turkey to 0.83 in Paraguay. Under domestic law it is at most 0.25, and six countries issue domestic-law bonds every quarter. Relative to that quarter's GDP, the median foreign-law operation is larger than the median domestic-law one in ten of the twelve countries, with a ratio that ranges from 1.1 in Poland to 90 in Uruguay; Brazil and Turkey are the two exceptions.

The unit of count is the operation: one day on which bonds settle in a market, with everything that settles that day added up. Re-openings of outstanding foreign-law bonds count as operations, because each is a new sale abroad with its own prospectus and fee. Both laws use the placements of the previous sections: Dealogic for foreign law and debt-office records for domestic law. We convert each operation to US dollars and divide it by that quarter's nominal GDP in US dollars.

Table 24. Issuance by Governing Law, Quarterly

	Inaction rate		Operations per quarter		Spacing,	One operation, % of quarterly GDP		
	Foreign (1)	Domestic (2)	Foreign (3)	Domestic (4)	quarters (5)	Foreign (6)	Domestic (7)	Ratio (8)
Brazil	0.54	0.00	0.61	20.1	2.1	0.22	0.48	0.5
Chile	0.75	0.17	0.36	3.4	3.1	2.51	0.45	5.5
Colombia	0.55	0.09	0.50	8.4	2.3	1.95	0.28	7.0
Hungary	0.62	0.00	0.51	10.0	2.6	3.38	0.58	5.9
Mexico	0.34	0.00	0.84	15.1	1.5	0.61	0.25	2.4
Paraguay	0.83	0.25	0.17	1.7	3.7	5.11	0.15	34.6
Peru	0.80	0.05	0.22	6.7	4.4	2.54	0.09	26.8
Poland	0.41	0.00	0.99	5.4	1.7	0.92	0.81	1.1
Russia	0.80	0.01	0.20	8.6	3.1	0.61	0.09	7.0
South Africa	0.80	0.00	0.21	19.5	5.3	1.87	0.19	9.7
Turkey	0.26	0.00	1.12	5.6	1.4	0.73	0.76	1.0
Uruguay	0.64	0.19	0.39	4.0	2.9	7.95	0.09	90.0

Notes: 2006Q1–2024Q4. An operation is one settlement day in a market, with amounts summed. Foreign-law placements from Dealogic, counting re-openings as operations; domestic-law placements from debt-office records, bonds only. Exchange operations are excluded on both sides. Columns (1)–(2) give the share of quarters with no issuance, (3)–(4) the operations per quarter, (5) the mean number of quarters between foreign-law issuance quarters, (6)–(7) the median operation as a percentage of that quarter's GDP, and (8) the ratio of (6) to (7).

A.14 Recovered debt in the numerical solution

We solve the model on a uniform grid for each bond. Domestic-law debt takes the values $b_D(i) = \bar{b}_D\, i/(N_D - 1)$ for $i \in \{0, 1, \ldots, N_D - 1\}$, and foreign-law debt is defined in the same way on N_F points. A government that regains access after a default carries $\nu_D b_D$ and $\nu_F b_F$. Because each grid is linear in its index and starts at zero, scaling the debt level by the recovery is the same as scaling the index: $\nu_D b_D(i) = b_D(\nu_D i)$. The recovered position $p_D = \nu_D i$ is generally not an integer, and the same holds for $p_F = \nu_F j$. Since recoveries are at most one, the recovered point never leaves the grid.

Each recovered position lies between two grid points, $i^- = \lfloor p_D \rfloor$ and $i^+ = \min(i^- + 1, N_D - 1)$, with weight $a = p_D - i^-$, and likewise j^-, j^+ and $c = p_F - j^-$ for foreign-law debt. We evaluate the new-issuance prices q_D^{ND} and q_F^{ND} and the continuation value at the recovered point by bilinear interpolation of the four surrounding grid values,

$$\tilde{f}(p_D, p_F) = (1-a)(1-c)\, f[i^-, j^-] + a(1-c)\, f[i^+, j^-] + (1-a)c\, f[i^-, j^+] + ac\, f[i^+, j^+]. \tag{36}$$

The weights are nonnegative and sum to one, and when a recovered position falls on a grid point the formula uses that point alone. Each law's recovery therefore scales its own

debt, and the recovered portfolio need not lie on the grid.

A.15 Depreciation at default and the calibration

To be added.

A.16 Robustness to the reentry probability

To be added.

A.17 The two experiments

Table 25. Same Durations, Different Recoveries: Foreign-Law Share (%)

	Duration of both bonds (years)				
ν_D / ν_F	1.00	2.00	3.75	5.50	8.71
0.200 / 0.400	100.0	100.0	100.0	100.0	100.0
0.400 / 0.420	100.0	100.0	100.0	100.0	100.0
0.400 / 0.471	100.0	100.0	100.0	100.0	100.0
0.400 / 0.550	100.0	100.0	100.0	100.0	100.0
0.400 / 0.700	100.0	100.0	100.0	100.0	100.0
0.471 / 0.400	0.0	0.0	0.0	0.0	0.0

Notes: Foreign-law share of debt when both bonds have the same duration. Other parameters at their calibrated values. The last row reverses which law has the higher recovery.

Table 26. Same Durations, Different Recoveries: Recovery Grid (%)

	ν_F				
ν_D	0.300	0.375	0.450	0.525	0.600
0.300	39.7	100.0	100.0	100.0	100.0
0.375	0.0	39.7	100.0	100.0	100.0
0.450	0.0	0.0	45.6	100.0	100.0
0.525	0.0	0.0	0.0	43.7	100.0
0.600	0.0	0.0	0.0	0.0	39.1

Notes: Foreign-law share of debt, both bonds with a duration of 3.75 years. On the diagonal the two bonds are identical and the split is not determined by the model.

Table 27. Same Recoveries, Different Durations: Foreign-Law Share (%)

	Duration, domestic / foreign law (years)		
$\nu_D = \nu_F$	1 / 8.71	2 / 8.71	3.75 / 8.71
0.00	47.8	2.0	0.3
0.40	58.8	2.0	0.3
0.4715	60.4	2.0	0.1
0.60	65.3	2.0	0.1
0.75	67.8	3.7	0.1

Notes: Foreign-law share of debt when both bonds have the same recovery. The last column uses the calibrated durations. Other parameters at their calibrated values.

A.18 Proof of Lemma 1

Events. Fix the government's policies and start at date 0 from a state $s_0 = (y_0, b_{D,0}, b_{F,0})$ at which one unit of each bond is issued. Along a path of output and reentry draws, the policies determine in each period $t \geq 1$ whether the government repays and whether it reenters:

$$\chi_t = \begin{cases} 1 & \text{repays at } t \\ 0 & \text{otherwise} \end{cases}, \qquad \eta_t = \begin{cases} 1 & \text{reenters at } t \\ 0 & \text{otherwise} \end{cases}, \qquad \chi_t \, \eta_t = 0.$$

The government reenters only from default, so it never repays and reenters in the same period. The events are the same for both bonds: the path of states follows from output and the policies, the government repays both bonds or defaults on both, and it reenters on all its debt at once.

Holdings and payments. Let $h_{j,t}$ be the units of law-j debt held at the start of t, before the events of t, by an investor who bought one unit at date 0. At a repayment a fraction λ matures, and at reentry each unit becomes ν_j units:

$$h_{j,1} = 1, \qquad h_{j,t+1} = h_{j,t} \times \begin{cases} 1 - \lambda & \chi_t = 1, \\ \nu_j & \eta_t = 1, \\ 1 & \text{otherwise.} \end{cases}$$

With $a_t \equiv \sum_{s<t} \chi_s$ repayments and $k_t \equiv \sum_{s<t} \eta_s$ reentries before t,

$$h_{j,t} = (1-\lambda)^{a_t} \, \nu_j^{k_t}, \qquad c_{j,t} = \chi_t \, \lambda \, h_{j,t} = \chi_t \, \lambda \, (1-\lambda)^{a_t} \, \nu_j^{k_t},$$

where $c_{j,t}$ is the payment at t: the matured fraction of the holdings if the government repays, and zero otherwise.

Prices as present values. Investors are risk neutral, so the price of a bond is the expected present value of its payments:

$$q_j^{ND}(s_0) = \mathbb{E}_0 \sum_{t=1}^{\infty} \frac{c_{j,t}}{(1+r)^t}. \tag{37}$$

Equations (11)–(14) are the recursive form of (37). The same expression gives $q_j^D(s_0)$, with the path starting in default at date 0 and reentry possible at date 0.

Comparison. By (37),

$$q_F^{ND}(s_0) - q_D^{ND}(s_0) = \mathbb{E}_0 \sum_{t=1}^{\infty} \frac{\chi_t \, \lambda \, (1-\lambda)^{a_t}}{(1+r)^t} \left(\nu_F^{k_t} - \nu_D^{k_t} \right) \;\geq\; 0,$$

since $\nu_F \geq \nu_D \geq 0$ gives $\nu_F^k \geq \nu_D^k$ for every $k \geq 0$. The same holds for $q_F^D - q_D^D$.

Strictness. If $\nu_F > \nu_D$, then $\nu_F^k > \nu_D^k$ for every $k \geq 1$, so each term with $\chi_t = 1$ and $k_t \geq 1$ is positive. The gap is positive whenever, with positive probability, the government reenters and then repays.

A.19 The fee and the cost of borrowing

This appendix shows why a fee charged once per unit of face value issued weighs more on short bonds than on long ones.

What the fee is charged on. The fee f is charged on face value, the amount the government promises to repay, and it is paid once, when the bond is issued. This is how the prospectuses of Section 2.5 quote it: an underwriting discount of 0.200 percent on US$2 billion of face is US$4 million, whatever the price at which the bonds are sold.

The bond. A bond with decay λ and no coupon pays, per unit of face, λ next period, $\lambda(1-\lambda)$ the period after, and so on: each period a share λ of the face still outstanding matures. Its yield y is the constant rate that discounts these payments to the bond's price q:

$$q = \sum_{k=1}^{\infty} \frac{\lambda(1-\lambda)^{k-1}}{(1+y)^k} = \frac{\lambda}{1+y} \sum_{k=0}^{\infty} \left(\frac{1-\lambda}{1+y} \right)^k = \frac{\lambda}{1+y} \cdot \frac{1+y}{y+\lambda} = \frac{\lambda}{y+\lambda}. \tag{38}$$

Solving for the yield gives $y = \lambda/q - \lambda$. A one-period bond has $\lambda = 1$ and $y = 1/q - 1$. For a given yield, the price rises with λ: a bond that repays sooner is discounted less, so short bonds trade closer to par than long bonds.

The fee as a lower price. With the fee, the government sells one unit of face for q but keeps only $q - f$, while it owes the same payments. The yield it pays on the money it actually receives is therefore

$$\hat{y} = \frac{\lambda}{q-f} - \lambda. \tag{39}$$

Subtracting y and writing the difference over a common denominator,

$$\hat{y} - y = \lambda \left(\frac{1}{q - f} - \frac{1}{q} \right) = \frac{\lambda f}{q\,(q - f)} \;\approx\; \frac{\lambda f}{q^2}, \tag{40}$$

where the approximation replaces $q - f$ by q, since the fee is small relative to the price. The price enters twice. Dividing the fee by q turns a cost per unit of face into a cost per unit of money borrowed, and dividing by q again turns a change in the price into a change in the yield, since $\partial y / \partial q = -\lambda / q^2$.

Short against long bonds. We compare the wedge on two bonds of the same government that differ only in their decay rate and trade at the same yield. This is the relevant comparison in the model: the two foreign-law bonds share the same recovery, so they carry the same mean spread and the same mean yield. A shorter bond has a larger λ, which raises the wedge $\lambda f / q^2$, but also a higher price q, which lowers it. To see which force dominates, substitute the price from equation (38), $q = \lambda / (y + \lambda)$, into the wedge, so that it depends on λ and y alone:

$$\frac{\lambda f}{q^2} = \lambda f \cdot \frac{(y + \lambda)^2}{\lambda^2} = \frac{f\,(y + \lambda)^2}{\lambda}. \tag{41}$$

At a given yield, the wedge changes with the decay rate according to

$$\frac{\partial}{\partial \lambda} \left[\frac{f\,(y + \lambda)^2}{\lambda} \right] = f\,\frac{2(y + \lambda)\lambda - (y + \lambda)^2}{\lambda^2} = f\,\frac{(y + \lambda)(\lambda - y)}{\lambda^2}, \tag{42}$$

which is positive whenever $\lambda > y$, that is, whenever the bond repays a larger share of its face each period than its yield. Every bond in the paper satisfies this condition: the slowest decay rate we use is $\lambda_F = 0.099$, the foreign-law bond of Section 4.1 with a duration of 8.71 years, and yields are a few percent. The wedge is therefore larger on the shorter bond. Its price is higher, but the price falls with maturity much less than λ does, so even squared it offsets only part of the difference in λ.

The table evaluates these expressions at the fee of 25 basis points and a common yield of 4.2 percent, the risk-free rate plus Colombia's mean spread. The first two rows are the short and the long foreign-law bonds of Section 5.1; the third is the foreign-law bond of Section 4.1. To keep one unit of face outstanding, the government issues λ units each period and pays $f\lambda$ in fees; the last column is the wedge in the yield.

Bond	λ	Price q	$f\lambda$ (bp of face)	Wedge $\lambda f/q^2$ (bp a year)
One period	1.000	0.960	25.0	27.1
2.32 years	0.420	0.909	10.5	12.7
8.71 years	0.099	0.702	2.5	5.0

From one period to 8.71 years, λ falls by a factor of ten and the price by less than a third, so the wedge falls by a factor of more than five. A higher yield on the long bond would lower its price and raise its wedge, but by equation (41) the 8.71-year bond would need a yield above 23 percent, and the 2.32-year bond a yield above 25 percent, for its wedge to reach that of the one-period bond at 4.2 percent.

A.20 Maturity under a fee on issuance

In a simple model without default, a fee proportional to the amount issued leads the government to issue only the longest bond.

Environment. Time is discrete and infinite, $t = 0, 1, 2, \ldots$, and there is no uncertainty. The government receives a constant endowment y and values consumption at $\sum_{t\geq 0} \beta^t u(c_t)$, with u strictly increasing and strictly concave. Lenders discount at the interest rate r, and $\beta = 1/(1+r)$. The government always repays. It starts period 0 owing $b > 0$, due in full in period 0, which it can repay out of its endowment or roll over.

At $t = 0$ the government chooses a decay rate $\xi \in [0, 1/\beta)$, and every bond it issues, at that date and later, has this decay rate. A unit issued at t pays 1 at $t+1$, ξ at $t+2$, ξ^2 at $t+3$, and so on. Lenders price it at

$$q(\xi) = \sum_{k\geq 1} \beta^k \xi^{k-1} = \frac{\beta}{1-\beta\xi}.$$

The one-period bond is $\xi = 0$ and the consol is $\xi = 1$.

Let i_t be the units issued at t, with $i_t < 0$ for a buyback. The government pays a fee $f \in [0, 1)$ on the market value of what it issues or buys back: issuing $i_t > 0$ units raises $(1-f)\, q\, i_t$, and buying back $|i_t|$ units costs $(1+f)\, q\, |i_t|$. Without the fee on buybacks the

consol remains optimal. With B_t the payment due at t, the budget constraints are

$$\begin{aligned} c_0 &= y - b + q\, i_0 - f\, q\, |i_0|, \\ c_t &= y - B_t + q\, i_t - f\, q\, |i_t|, \qquad t \geq 1, \\ B_1 &= i_0, \qquad B_{t+1} = \xi B_t + i_t, \qquad t \geq 1, \end{aligned}$$

and debt is bounded, $B_{t+1} \leq \bar{B}$ for all $t \geq 0$, with $\bar{B} \geq \max\{b,\ (1-\beta)\, b/((1-f)\,\beta)\}$.

Proposition 1 (Maturity under a fee)**.** *If $f = 0$, every ξ gives the same allocation, $c_t = y - (1-\beta)\, b$ for all t. If $f \in (0,1)$ and $(1-f)\, u'(y-b) > u'(y)$, the consol, $\xi = 1$, is the unique optimal maturity. The government issues only at $t = 0$, and consumption is c_0 at $t = 0$ and $\bar{c}$ at every $t \geq 1$, with*

$$(1-f)\, u'(c_0) = u'(\bar{c}), \qquad c_0 < \bar{c}.$$

If $(1-f)\, u'(y-b) \leq u'(y)$, the government does not borrow and maturity is irrelevant: $c_0 = y - b$ and $c_t = y$ for $t \geq 1$.

The fee is paid on every issue, so the government gains from a bond that it never has to roll over. With a constant endowment, the consol is the only bond whose payments match a flat path of consumption after date 0.

A.21 The fee ladder

Table 28. Shares of Debt and the Foreign-Law Fee

	Two bonds	Three bonds		
Fee (bp of face)	Foreign law	Short domestic	Short foreign	Long foreign
0	100.0	0.0	35.9	64.1
20	74.8	12.1	5.9	82.0
25 (Colombia)	—	13.2	3.0	83.8
35	—	18.5	0.7	80.7
Fee at which issuance stops (bp)				
Foreign law, two bonds	76			
Short foreign law, three bonds			35	

Notes: Shares of total face value, in percent, means over simulated periods. Two bonds: a one-period domestic-law bond and a one-period foreign-law bond. Three bonds: one-period domestic-law and foreign-law bonds and a foreign-law bond with a duration of 2.32 years. Issuance stops when the share falls below 1 percent. Other parameters as in Table 7.

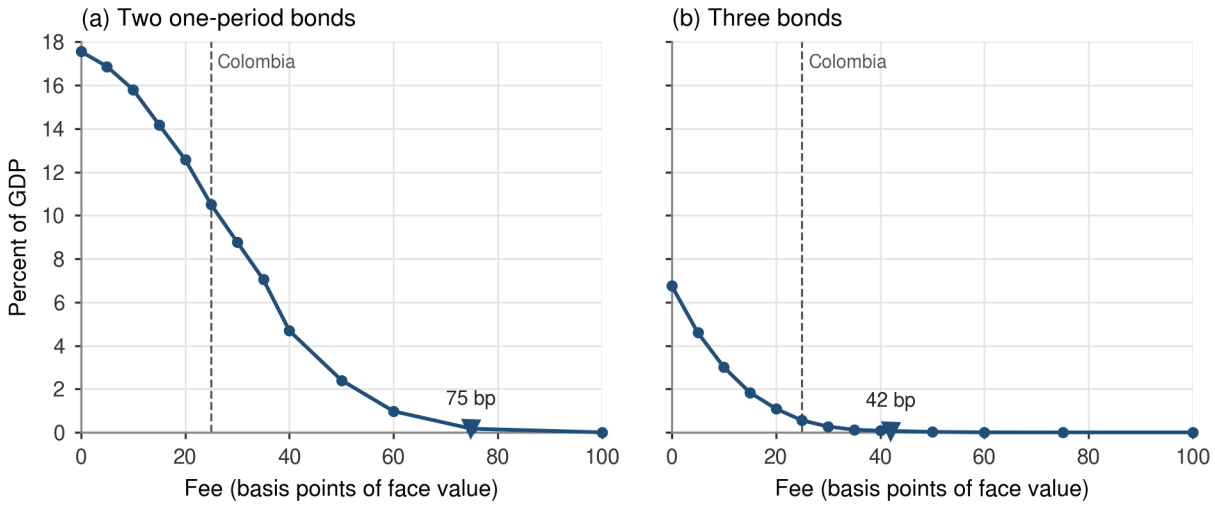


Figure 10. Foreign-Law Debt and the Fee

Notes: Mean debt in percent of GDP against the fee per unit of face value issued under foreign law. Panel (a): foreign-law debt when the government has a one-period domestic-law bond and a one-period foreign-law bond. Panel (b): short foreign-law debt when the government also has a foreign-law bond with a duration of 2.32 years. The dashed line marks Colombia's fee of 25 basis points. The triangle marks the fee at which the plotted debt falls below 1 percent of its value without a fee.

A.22 Welfare: measurement and decomposition

Decomposition with policies fixed. The price of defaulted debt,

$$q_i^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\,\mathbb{E}_{y'|y}\left[q_i^D(y', b_D, b_F)\right] + \theta\,\nu_i\,q_i^{ND}(y, \nu_D b_D, \nu_F b_F)\,,$$

shows the two roles of the recovery: a higher ν_i raises the price investors pay, and it raises the debt the government carries at reentry. Holding the borrowing and default policies at their baseline values, we compute welfare in four economies: the baseline; new prices with the baseline debt at reentry; baseline prices with the new debt at reentry; and both changes together. The first comparison isolates the price effect and the second the reentry effect. The difference between the economy with $\nu_D = 0.47$, in which policies adjust, and the economy with both changes and fixed policies is the effect of the government's behaviour.

The value of market access. For an economy $\mathcal{E}$, let $\bar{V}^{\mathcal{E}}$ be the household's expected discounted utility along the equilibrium path, with consumption $y - \phi(y)$ while the government is excluded. Let $\bar{V}^{ND,\mathcal{E}}$ be the same value when exclusion carries no output loss, so that consumption equals y while excluded and the government still reenters owing $\nu_D b_D$ and $\nu_F b_F$. Let $\bar{V}(\bar{c}^{ND,\mathcal{E}}) = \sum_t \beta_H^t u(\bar{c}_t^{ND,\mathcal{E}})$ be the value of receiving for sure the

mean path $\bar{c}_t^{ND,\mathcal{E}}$ of that consumption. Let $\bar{V}^A$ be the value of consuming output forever, and $\bar{V}^A(y^\infty)$ the value of consuming mean output $y^\infty = \sum_y \pi(y)\, y$ forever. Following Aguiar et al. (2020), the consumption-equivalent value of market access is

$$1+\chi^{\mathcal{E}} \equiv \left(\frac{\bar{V}^{\mathcal{E}}}{\bar{V}^A}\right)^{\frac{1}{1-\sigma}} = \underbrace{\left(\frac{\bar{V}^{\mathcal{E}}}{\bar{V}^{ND,\mathcal{E}}}\right)^{\frac{1}{1-\sigma}}}_{\text{default costs}} \underbrace{\left(\frac{\bar{V}^{ND,\mathcal{E}}}{\bar{V}(\bar{c}^{ND,\mathcal{E}})} \cdot \frac{\bar{V}^A(y^\infty)}{\bar{V}^A}\right)^{\frac{1}{1-\sigma}}}_{\text{variability}} \underbrace{\left(\frac{\bar{V}(\bar{c}^{ND,\mathcal{E}})}{\bar{V}^A(y^\infty)}\right)^{\frac{1}{1-\sigma}}}_{\text{front-loading}}. \quad (43)$$

The value of autarky is the same in the two economies, so the welfare gain of equation (19) is $1+\Delta = (1+\chi^{\mathcal{E}_1})/(1+\chi^{\mathcal{E}_0})$, and each term contributes the ratio of its own values across the two economies.